

A Structure-Aware Hierarchical Graph Neural Network for Multi-Label Scent Profile Prediction in Binary Fragrance Mixtures

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Senior Thesis submitted to the faculty of the
Department of Computer Science
College of Computer Studies, Ateneo de Naga University
in partial fulfillment of the requirements for their respective
Bachelor of Science degrees

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The Senior Project entitled

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We declare that the Senior Project entitled

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ABSTRACT

Perfumers and flavor chemists routinely combine multiple aroma chemicals into a single fragrance, yet the resulting scent cannot be reliably predicted from its individual components: molecular blending can suppress notes present in one or both parent molecules, or produce entirely new notes that neither molecule exhibits alone, phenomena documented in large-scale blend datasets. Existing computational approaches to olfactory prediction, including graph neural networks that achieve near-human accuracy on single molecules, do not model molecule-to-molecule interaction directly, while the few models that address molecular blends combine representations through simple pooling or joint-graph encoding, lacking an explicit interaction step. This study proposes a Structure-Aware Hierarchical Graph Neural Network (SA-HGNN) that predicts the multi-label scent of two-molecule fragrance blends directly from molecular structure. The architecture operates in two levels: a Graph Attention Network first encodes each molecule independently from its atomic and bond-level features into a fixed-length embedding, after which a Set Transformer, using self-attention and pooling by multihead attention, lets the two molecule embeddings interact before producing a blend-level prediction across a standardized vocabulary of scent descriptors. The model will be trained on molecular and blend data compiled from GoodScents, Leffingwell, and a published aroma-chemical blend dataset, evaluated on a leakage-resistant, structurally out-of-distribution split, and compared against pooling-based and reproduced pairwise-model baselines using macro F1, micro F1, PR-AUC, and Hamming loss, with statistical significance assessed through paired bootstrap resampling. The study further examines whether the model’s learned attention aligns with documented receptor-level antagonism. By reducing dependence on physical trial-and-error formulation, the proposed system

is intended to support more sustainable and innovative practice in fragrance development, in line with UN Sustainable Development Goals 12 and 9

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Chapter 1

Introduction

Computational chemistry faces a persistent challenge in modeling complex molecular interactions to predict odor descriptors, particularly when relying on architectures that are not inherently permutation-invariant [26, 44, 54]. Numerous studies have been conducted and successfully demonstrated that single-molecule graph neural networks have advanced property prediction for individual molecules, but real-world applications involve chemical blends. [40, 25]. However, standard machine learning models use simple linear pooling, such as mean or sum aggregation. Theoretically, using these naive methods of pooling may not fully capture receptor-level olfactory perception, where scents emerge and are masked [44]. This study will implement a Structure-Aware Hierarchical Graph Neural Network (SA-HGNN) by combining a level 1 Graph Attention Network (GAT) to encode individual molecules [52] with a level 2 Set Transformer to capture molecular interactions [26]. One of our goals is to preserve the symmetrical feature of chemical mixtures and remain permutation-invariant while still accounting for nonlinear interactions [54]. Ultimately, this research will establish a mathematical model for a binary-molecule blend prediction and a sustainable aroma-chemical substitution [30].

1.1 Project Context

Modern fragrance recommendations rely on fragrance communities, on which descriptors are inconsistently labeled on commercial perfumes. Established fragrance communities and scent repositories such as Basenotes and Fragrantica have become the source of ground truth for fragrance reviews,

olfactory descriptions, and everyday aesthetic discourse. The narratives are subjective and mostly labeled by untrained consumers. Scent descriptors that were used in these platforms are interchangeable due to branding preferences, which can create confusion among multiple fragrance descriptors that smell alike. This treats perfumes and fragrance mixtures as subjective descriptive words rather than physical chemical systems. These raw scent descriptions are why there is a need for a computational molecular analysis for an objective ground truth [6, 49].

Computational olfaction solves these limitations through Geometric Deep Learning by adopting modeling structure-odor relationships (SOR) directly from the nominal chemical structures of chemical compounds. Because molecules are inherently non-Euclidean topological structures, this framework represents individual aroma-chemicals as attributed molecular graphs ($G = (V, E)$), where atoms serve as feature-attributed nodes V and chemical bonds serve as edge attributes E . Graph Neural Networks (GNN) frameworks have become the cutting edge for learning representations of molecular graphs, as localized message-passing algorithms enable end-to-end learning of topology-aware olfactory features [31]. A structure to perception study has successfully modeled GNN on expertly curated single-molecule datasets, mapping molecular structure to human perception. The study demonstrated that their model’s prediction of a new molecule is closer to the mean of the 15-member human panel [25]. However, while single-molecule GNNs excel at single-graph classification, single-molecule GNNs are not designed to model a multi-graph set representation. This is the reason why it limits the capability to capture the perceptual and physical dynamics of multi-molecule mixtures, such as masking and emergence [46].

Scents are typically blends with multiple components rather than single compounds. When two aroma chemicals get mixed, the result isn’t just additive; it triggers complex psychophysical and physical interactions. In a study conducted by Sisson et al., they demonstrated that in an unweighted binary mixture, blends exhibited psychophysical distortions, losing an average of 1.2 constituent notes to suppression and gaining 0.2 novel descriptors through configural emergence [40]. Current machine learning models struggle to implement a solution for capturing these biological distortions, largely because they rely on naive linear aggregation, such as mean or sum pooling [54]. In deep learning theory, aggregating unordered sets of molecular vectors requires enforcing strict permutation invariance. However, linear pooling operations act as a group-averaging Reynolds projection, which flattens the multi-dimensional feature space—a phenomenon known as “symmetry collapse”. This imposes a gap that Standard Graph Neural Networks are limited in capturing non-linear interactions

that occur at the receptor level [44].

1.2 Purpose and Description

This study aims to build a computational molecular analysis model that uses nominal chemical structures rather than subjective scent descriptions. This study focuses on predicting the emergent, blend-level odor descriptors of binary (two-molecule) mixtures using hierarchical Graph Neural Networks (GNN) with Set Transformers [40]. The hierarchy is divided into three computational parts with two interaction layers and a classification head.

The level 1 consists of the Intra-Molecular GAT Encoder. A Graph Attention Network (GAT) encodes individual aroma-chemical topologies, producing a molecular representation for each molecule based on its own node attributes and covalent bond features [52]. At this first level, the molecular encoder uses deep message-passing layers to approximate the mapping function that maps molecular graphs to isolated, elemental scent descriptors.

The level 2 serves as the Inter-Molecular Set Aggregator. This extends the single-molecule GNN architecture into our SA-HGNN (Structure-Aware Hierarchical Graph Neural Network) framework using permutation-invariant attention enhancements [26, 54]. These enhancements consist of Self-Attention Blocks (SAB) and Pooling by Multihead Attention (PMA). Within this layer, the PMA module processes the multi-molecule embedding sets generated by level 1 and produces learned and interactive blend embeddings for the implicit inference of suppressed and emergent descriptors [26].

The Multi-Label Classification Head acts as the determining and output layer. This block utilizes a dense feedforward neural network to map the joint molecule representation to independent sigmoid activations, which will yield multi-label scent predictions.

The scent classification model is intended to benefit the following entities:

- **Fragrance Chemists and Perfumers.** This will provide them with an objective, structure-grounded tool that systematically models how chemicals interact in a blend. This will reduce their cost and save resources from trial-and-error cycles during perfume formulation [35].
- **Green Chemical Engineers.** This study provides a sustainable way to find safe and eco-friendly substitutes for regulated and endangered natural ingredients. The model allows engineers to design sustainable fragrances without threatening natural ecosystems through the mathematical matching of similar-smelling molecules [30].

- **Chemoinformatics and Deep Learning Researchers.** This study contributes a standardized framework for a two-molecule mixture prediction that could serve as a baseline for set-structured graph representations. This will also provide empirical evidence on whether attention-based set decoders overcome the “symmetry collapse” limitation of linear graph readouts [44].

Research Questions

This study seeks to answer the following questions:

1. To what extent does the Level 2 Set Transformer interaction layer improve blend odor descriptor prediction over naive pooling baselines and the reproduced pairwise blend model, and what does the magnitude of this gap reveal about whether odor mixing behaves additively at this representational level?
2. How much does the best-performing model’s performance degrade on structurally novel aroma-chemical scaffolds under a leakage-resistant clustering split (UMAP), compared to a standard random split?
3. How well does the model’s learned attention distribution align with documented receptor-level antagonism in binary mixtures where one molecule is known to mask another, and what patterns of agreement or divergence does this reveal?

1.3 Objectives

General Objective

To design and evaluate a Structure-Aware Hierarchical Graph Neural Network (SA-HGNN) model that pairs a single-molecule Graph Attention Network (GAT) with a Set Transformer interaction layer to predict multi-label scent profiles in binary fragrance blends, capturing both suppressed and emergent descriptors. To achieve the general objective, the following specific objectives must be accomplished:

- To preprocess the single-molecule datasets from GoodScents and Leffingwell, and pair them with the binary mixture scent records.

- To build the Level 1 Graph Attention Network (GAT) that translates individual chemical structures, specifically their atoms and bonds, into uniform scent vectors.
- To pre-train the Level 1 Graph Attention Network on individual compounds, so the model learns basic structure-to-odor relationships before it is evaluated on mixtures.
- To design and integrate the Level 2 Inter-Molecular Set Aggregator using a Set Transformer by incorporating Self-Attention Blocks (SAB) and Pooling by Multihead Attention (PMA) to model dynamic cross-molecule interactions such as suppression and emergence.
- To implement a Multi-Label Classification Head that maps the combined pairs from the Level 2 Set Transformer to independent sigmoid activations, predicting the presence of up to 74 distinct, expert-assigned scent notes, with the final count determined by the survival filter after graph carving.
- To organize chemical structures into families using UMAP clustering on molecular fingerprints, to test whether the model predicts unfamiliar chemical groups.
- To execute end-to-end joint fine-tuning of the combined SA-HGNN pipeline on binary mixtures, utilizing asymmetric loss functions to mitigate label imbalance and high sparsity.

1.4 Scope and Limitations

This study will focus on the design, development, and evaluation of a Structure-Aware Hierarchical Graph Neural Network (SA-HGNN). The primary scope of this network is restricted to pairwise or two-molecule blends. The framework is designed to predict the multi-label scent profiles of binary fragrance mixtures that will utilize a two-level architecture to capture complex blending dynamics rather than relying on simple molecular averaging or naive pooling [40, 44]. The first layer of the network is a Level 1 Intra-Molecular Graph Attention Network (GAT) encoder to process 2D molecular graphs [52]. The second network layer is a Level 2 Inter-Molecular Set Transformer interaction layer that will utilize a Self-Attention Block (SAB) and Pooling by Multihead Attention to model non-linear blending dynamics [26, 54].

The dataset pipeline merges single-molecule structural and olfactory data from the GoodScents and Leffingwell to be extracted from the Pyrfume library in Python [6]. Alongside the merged

data is a benchmark dataset of expert-labeled binary chemical blends from Sisson et al.’s study. Model optimization involves two-phase training protocols. The first phase includes pre-training the Level 1 GAT on individual aroma-chemicals to learn structural-olfactory baselines. The second phase executes end-to-end joint fine-tuning on the binary mixtures. The optimization will utilize an Asymmetric Loss function to mitigate the severe label sparsity and class imbalance, which are known to be in multi-label olfactory data [17].

A Leakage-resistant evaluation split will be implemented in evaluating the model’s out-of-distribution (OOD) generalization [11]. The chemical space will be clustered into groups and families using UMAP dimensionality reduction and K-Means clustering on UMAP Projection of 2048-bit Morgan fingerprints (ECFP4), followed by K-Means clustering on the 2D coordinates [14]. The performance results will be compared with standard averaging baseline—mean pooling—and will be replicated in existing pairwise models across Hamming Loss, Macro and Micro F1-scores, and Area Under the Precision-Recall Curve (PR-AUC). The attention weights in Level 2 will be extracted and correlated with known olfactory interactions, specifically masking and suppression, which are documented in psychophysical literature [20].

Attention-versus-Antagonism Validation Protocol. To operationalize Research Question 3, this study evaluates whether the attention distribution learned by the Level 2 Set Transformer reflects biological receptor-level antagonism and perceptual masking. Candidate antagonist–agonist pairs with documented in vitro or psychophysical receptor suppression are compiled from benchmark literature: β -ionone blocking OR2T receptors activated by sulfide malodors (CH_3SH , H_2S) [12], L-menthol blocking OR5K1 activated by 2,3,5-trimethylpyrazine [20], and Florhydral blocking OR51E1 activated by isovaleric acid [21]. These pairs are cross-referenced against the single-molecule master dictionary and Sisson et al.’s [40] binary blend dataset via PubChem CIDs and canonical SMILES to verify their presence in the evaluation partitions.

For candidate literature pairs present in the dataset, a successful alignment is operationalized by attention weight asymmetry, where the documented blocking odorant receives a higher Pooling by Multihead Attention (PMA) weight than the antagonized agonist ($\alpha_{\text{blocker}} > \alpha_{\text{agonist}}$) alongside attenuation of the agonist’s primary descriptor. If candidate physiological pairs are absent because Sisson et al.’s corpus reflects commercial fragrance formulations rather than synthetic antagonist bioassays, a systematic fallback evaluates global perceptual suppression: mixture pairs exhibiting note suppression (extinguished component descriptors) are compared against purely additive pairs.

The PMA attention skew $|\alpha_1 - \alpha_2|$ and attention entropy are compared across both groups to test whether suppressed pairs exhibit significantly higher attention asymmetry toward the dominant odorant compared to the balanced attention ($\alpha_1 \approx \alpha_2 \approx 0.5$) expected in additive mixtures.

1.5 Definition of Terms

Because this study operates at the intersection of chemosensory science and geometric deep learning, key domain concepts, representations, and evaluation protocols are operationalized as follows to establish an accessible foundation across both fields:

Aroma Molecule to Attributed Molecular Graph

In fragrance chemistry, an aroma molecule is a volatile organic compound that activates human olfactory receptors. In this study, each molecule is computationally represented as an undirected, attributed molecular graph $G = (V, E)$, where vertices V denote non-hydrogen heavy atoms encoded with a 46-dimensional feature vector capturing elemental identity, valence, hybridization, aromaticity, and chirality, and edges E denote covalent bonds encoded with a 10-dimensional feature vector.

Scent Profile to Multi-Label Binary Vector

The sensory character of a fragrance is defined by expert-assigned odor descriptors (e.g., woody, floral, citrus). This qualitative profile is operationalized as a multi-label classification target vector $\mathbf{y} \in \{0, 1\}^L$, where each binary dimension indicates the presence or absence of a specific descriptor from a standardized vocabulary of up to $L = 74$ notes.

Binary Fragrance Blend to Permutation-Invariant Set

A binary mixture consists of two distinct aroma chemicals combined without a chemical reaction. Computationally, this mixture is formalized as an unordered multiset of latent molecular embeddings $\{\mathbf{z}_1, \mathbf{z}_2\}$. Because physical odor perception is invariant to the presentation order of the components, the blend-level aggregation function must strictly satisfy permutation invariance.

Receptor Antagonism and Masking to Asymmetric Cross-Attention

Biologically, antagonism occurs when one odorant binds to an olfactory receptor and blocks an agonist from activating it, causing perceptual masking or suppression of constituent notes. In the proposed SA-HGNN architecture, this non-linear interaction is mapped to the Pooling by Multi-

head Attention (PMA) mechanism of the Level 2 Set Transformer, where the model dynamically learns asymmetric attention weights ($\alpha_{\text{dominant}} > \alpha_{\text{suppressed}}$) rather than enforcing uniform linear averaging.

Constituent Suppression and Emergence to Non-Linear Label Transitions

Psychophysical blend phenomena include note suppression (the loss of a constituent note present in a single-molecule parent) and configural emergence (the appearance of a novel descriptor absent in both parents). In this investigation, these phenomena represent non-linear label transitions that serve as the empirical basis for demonstrating that attention-driven set readouts outperform linear pooling baselines.

Sustainable Substitute Screening to Latent Metric-Space Retrieval

Under international regulatory standards (IFRA) and UN Sustainable Development Goal 12, fragrance reformulation requires identifying non-toxic, eco-friendly alternatives for hazardous or restricted fragrance ingredients. This process is operationalized as a k -nearest-neighbor retrieval task in the learned Level 1 GAT latent space ($\mathbb{R}^{256}$), identifying candidate molecules that maximize cosine similarity to preserve target odor character while satisfying environmental and safety constraints.

SMILES Representation

The Simplified Molecular Input Line Entry System (SMILES) is a standardized text notation that encodes 2D molecular structures into compact ASCII strings. In this pipeline, raw SMILES strings serve as the primary computational input format, parsed and chemically sanitized via the RDKit library into attributed molecular graphs.

Morgan Fingerprints (ECFP4)

Extended-Connectivity Fingerprints (ECFP4) are 2048-bit binary vectors that record the presence or absence of circular chemical sub-structures around each atom up to a radius of two bonds. In this study, Morgan fingerprints provide an independent structural baseline used to cluster molecules into distinct chemical families via UMAP and K -means for evaluation splits.

Graph Attention Network (Level 1 GAT)

A geometric deep learning architecture that updates atomic feature vectors by computing dynamic attention weights across neighboring covalent bonds. In Level 1 encoding, GAT enables the network

to focus on the most salient functional groups (e.g., esters or aromatic rings) when projecting a molecule into a 256-dimensional embedding.

Set Transformer (Level 2 Aggregation)

An attention-based neural network designed specifically for processing unordered sets of feature vectors. By combining a Self-Attention Block (SAB) to model pairwise relationships between constituent embeddings and Pooling by Multihead Attention (PMA) to generate a fixed-size blend representation, it captures intermolecular interactions while preserving permutation invariance.

Leakage-Resistant Evaluation Split

A data partition protocol designed to prevent overly optimistic evaluation caused by data leakage. Instead of splitting mixtures uniformly at random, molecules from the same structural cluster are held out together, testing whether the model can generalize accurately to unfamiliar, out-of-distribution (OOD) chemical families.

Asymmetric Loss (ASL)

A specialized multi-label loss function engineered for extreme class imbalance and label sparsity. By dynamically shifting and down-weighting the loss contributions of easily classified negative labels, ASL prevents the model from being overwhelmed by the thousands of absent scent notes, prioritizing the accurate detection of the few positive descriptors present.

Hamming Loss

A standard multi-label evaluation metric that measures the fraction of misclassified instance-label pairs. In scent profiling, it quantifies the average rate of false positives (hallucinated notes) and false negatives (missed notes) across all L possible descriptors, where lower values indicate superior overall multi-label accuracy.

Cheminformatics Sanitization (RDKit)

The automated validation pipeline that enforces physical and chemical rules on parsed molecular graphs. It executes kekulization, aromaticity perception, valence verification, and ring system identification to eliminate syntactically invalid or hypervalent structures prior to deep learning ingestion.

Chirality and Stereochemical Tags

The spatial geometry of atoms exhibiting non-superimposable mirror-image forms (enantiomers) that often trigger distinct olfactory receptors. In this study, chirality is explicitly preserved through 4-dimensional tetrahedral stereocenter tags embedded directly within the 46-dimensional atomic node vectors.

Principal Odor Map (POM)

A continuous, data-driven coordinate space wherein geometric distances correspond to perceptual differences between aroma chemicals. Rather than treating scent descriptors as isolated labels, the POM paradigm projects molecules onto an organized perceptual manifold.

Message Passing Paradigm

The algorithmic framework of Graph Neural Networks wherein atom nodes iteratively update their internal feature states by aggregating localized messages from bonded neighbors, enabling the network to learn rich sub-structural chemical representations.

Self-Attention Block (SAB) and Pooling by Multihead Attention (PMA)

The two functional building blocks of the Level 2 Set Transformer. SAB performs all-to-all cross-attention among constituent molecules to model pairwise interaction dynamics, while PMA uses learned seed query vectors to compress the resulting set into a fixed-length blend embedding.

Symmetry Collapse

The mathematical degradation that occurs when linear set operations (such as mean or sum pooling) project multidimensional feature vectors onto a uniform Reynolds average, destroying asymmetric relational information required to distinguish complex mixtures.

Macro and Micro F1-Scores

Evaluation metrics balancing precision and recall in multi-label classification. Macro F1 averages performance independently across each scent descriptor to evaluate accuracy on rare notes, whereas Micro F1 aggregates true positives, false positives, and false negatives globally across all predictions.

Area Under the Precision-Recall Curve (PR-AUC)

A performance metric that calculates the area beneath the precision-recall trajectory across various classification thresholds, serving as the gold standard for evaluating model reliability under severe

class imbalance and label sparsity.

Chapter 2

Review of Related Literature

This chapter discusses the relevant studies and systems that support this thesis. First, it tackles seminal research on computational olfaction, followed by the science behind the perception of scents and how different scents behave when mixed. It also examines the application of machine learning to scent prediction when the molecular components in a blend are viewed as a set. Next, the chapter explains the biological architecture beneath these interactions. It reviews hierarchical Graph Neural Networks (GNNs) and their use in the computational modeling of these biological and chemical processes, and presents the literature that justifies the proposed evaluation methodology. Furthermore, it contextualizes the justification for this research, specifically its contribution to sustainable fragrance design, given the ongoing challenges in the domain. This chapter concludes by synthesizing the information from the gathered literature to identify the specific research gap this thesis addresses.

2.1 Computational Olfaction and Single-Molecule Prediction

This first section explores the evolution of computational olfaction from early QSOR modeling with manual feature engineering to more modern graph-based methods. It reviews how these frameworks were used in predicting scent from a single molecule and the limitations of such when applied to the multi-molecule blend prediction task this thesis aims to achieve.

2.1.1 History of QSOR Modeling

Quantitative Structure-Odor Relationship (QSOR) modeling refers to the prediction of the odors and olfactory properties of molecules through the characterization of their structure [13, 50, 55]. It uses the application of chemistry, neuroscience, biology, and psychology to uncover the mechanics behind odor perception, and has long been regarded as one of the key challenges in olfactory research in machine learning [6, 29, 52]. QSOR is especially important in fields such as the fragrance and flavor industries, as it could help streamline the formulation and development of new compounds computationally, as opposed to traditional experimental wet lab methods, which can be both time-consuming and expensive [18, 37].

Early approaches in QSOR involved the manual creation of numerical features that describe and quantify the properties of a molecule [3, 13, 56]. This typically meant either molecular fingerprints, which represent a molecule by encoding the presence or absence of specific substructures into a binary vector [3, 10, 38, 49], or physicochemical descriptors. Examples of fingerprints include Daylight fingerprints, developed by Daylight Chemical Information Systems, which use a path-based featurization approach wherein the connectivity of molecular bonds is analyzed. On the other hand, Morgan or extended-connectivity fingerprints (ECFP) are more modern; instead of tracing linear paths, they take a circular approach, capturing neighboring atoms iteratively within a defined radius [4, 10, 28]. Aside from fingerprints, molecules can also be numerically described using physicochemical descriptors, which characterize molecular structures by their topological, geometric, electronic, and thermodynamic features, computed using software such as Dragon and Mordred [3, 38].

Suh et al., for instance, compared classical molecular descriptors, functional-group fingerprints, and Morgan fingerprints across random forest, XGBoost, and LightGBM on a curated set of 8,681 fragrance compounds, finding that the Morgan-fingerprint models outperformed the descriptor-based ones [43]. Achebouche et al. predicted both the odors of molecules and the olfactory receptors they activate using a set of 5,955 molecules, and found that graph convolutional representations outperformed general molecular descriptors and fingerprints [1]. Saini and Ramanathan demonstrated the use of both Morgan fingerprints and Mordred descriptors, paired with a random forest classifier with multi-label support, to predict odor from molecular structures [37]. Wen et al. also used approximately 5,000 Dragon descriptors with RF, SVM, gradient boosting decision trees, multilayer perceptron, XGBoost, and LightGBM, comparing them against GNN approaches to also predict odor [48].

The common denominator across all of these approaches is that they convert the molecule, specifically its properties, into fixed, handcrafted numerical representations, which are then fed into classical ML models. Recent work such as Suh et al., Achebouché et al., and Wen et al. continues to include descriptor-based classical models, but mainly as baselines to quantify how much fingerprint- and graph-based learning improves upon manual feature engineering [1, 43, 48]. While effective, these methods across both eras require the manual selection of descriptors in advance [4, 25, 56]. This means the model is not actually able to determine on its own which structural features are relevant for the task it is trying to solve. This necessitated the advancement to graph-based deep learning methods, which improve the odor prediction pipeline by teaching models to learn task-relevant features directly from structure [2, 4, 16, 55] which will be discussed in the next section.

2.1.2 The Principal Odor Map

Addressing the aforementioned limitation is Lee et al.’s Principal Odor Map (POM), which maps a molecule’s chemical structure to its smell using Graph Neural Network (GNN) techniques [25]. Rather than evaluating a molecule’s scent using a static representation built from predefined handcrafted descriptors, POM learns directly from the molecule’s structure during training. It does this by treating each molecule as a graph, where nodes and edges represent atoms and chemical bonds, respectively; each node and edge is itself a vector, with nodes storing properties such as atomic number, and edges storing bond properties.

Through message passing, the GNN dynamically assigns weights to each of the molecule’s structural features across multiple passes, producing a numerical representation of the molecule. The final layer predicts the molecule’s specific odor qualities, while the penultimate layer forms the actual Principal Odor Map, which the paper itself describes as a high-dimensional, data-driven representation that is able to preserve human perceptual distances and hierarchies, positioning molecules with similar smells closer together in the generated odor space [25].

The setup was as follows: Lee et al. used approximately 5,000 molecules derived from merging data from the GoodScents and Leffingwell fragrance databases to produce a multi-label output of 138 distinct semantic odor descriptors. It was then validated against a human benchmark, wherein approximately 15 trained panelists rated molecules using a narrowed-down set of 55 odor descriptors. The results show it was able to match the trained panel’s verdicts on every label except for musky scents, and was even able to surpass the median panelist on 30 out of the 55 validation labels (55%)

[25].

This demonstrates the relative effectiveness of GNNs in predicting the scent of a molecule from structure alone. However, despite this significant advancement, these results were derived exclusively from single, unmixed molecules.

2.1.3 Limitations of Single-Molecule Models

Despite GNNs like POM being able to achieve human-like performance in scent prediction, the fact that they only operate on the scale of a single molecule demonstrates a fundamentally limited architecture that fails to reflect the reality that odorants rarely appear in isolation [8, 25, 46, 53]. Real-world scents are rarely composed of single molecules acting on single receptors; rather, they are blends of molecules that can contain individual components numbering up to the hundreds [15, 53]. Consequently, when a method focuses only on a single isolated graph, it is unable to predict new scents that may emerge from the mixture of two or more compounds.

Lee et al. admit this limitation themselves, noting that while POM uses a coordinate system to categorize scents, it alone cannot determine the behavior and outcomes of different molecules when mixed [25]. Moreover, single-molecule coordinate systems do not reflect a biologically plausible architecture, failing to capture the biological complexity of how the mammalian peripheral olfactory system actually operates [7, 19].

2.2 Psychophysics of Odor Mixtures

The unsuitability of single-molecule models for odor prediction in multi-molecular blends signifies the need for a similarly multi-molecular model. This section discusses the groundwork for predicting the behavior of different molecules when mixed, specifically the application of psychophysics to machine learning: how blends deviate from their original descriptors, how the brain processes them, and the concept of linearity in odor perception that will later justify the need for a Set Transformer instead of naive pooling.

2.2.1 Scent Blending Mechanics

A paper released by Sisson et al. introduces a labeled dataset consisting of molecule pairs to measure the divergence of the emerging scent when two molecules are mixed from their original scent labels,

using two metrics: union and intersection. The union score operates under the assumption that the emerging scent will smell like any of the scents either parent molecule has (additive mixing). On the other hand, the intersection score operates under the assumption that the outcome of the two molecules when mixed will only come from the scents the parent molecules share (subtractive mixing) [40].

The Jaccard results show 0.12 for union and 0.24 for intersection [40]. These low scores indicate that neither additive nor subtractive mixing reflects the real behavior exhibited by molecule pairs when mixed, according to the human-reported blend labels. This shows that the naive use of linear math cannot properly replicate the unpredictable behavior that molecule pairs exhibit when mixed.

This non-linearity can be characterized by the phenomena of masking/suppression and emergence, wherein the former describes what happens when a scent present in either parent molecule fails to appear in the blend, while the latter describes what happens when a new scent is perceived that was not present in either parent molecule’s odor descriptors. The results of Sisson et al. show that, on average, 1.2 notes are suppressed and 0.2 notes emerge per blend. Furthermore, an odds-ratio analysis reveals that certain scents, such as "eggy" or "cabbage", were more likely to emerge, while certain scents like "peach" or "fresh" were more likely to be suppressed [40]. This demonstrates that these two phenomena follow a certain learnable pattern, and that the behavior is not merely random, which motivates using a model that can learn these interactions, such as a Set Transformer.

2.2.2 Elemental vs. Configural Processing

Following the examination of the non-linear behavior of the scents of molecule pairs when mixed, this section will provide a sort of basis through the lens of how humans process odors. Studies by Romagny et al. and Coureaud et al. describe two strategies that the human brain uses to process smells, namely configural odour perception and elemental odour perception. With elemental processing, the brain distinguishes the different components that make up a scent. On the other hand, with configural processing, the brain perceives a new scent that is distinct from the original scents that make up its components [9, 36].

To better illustrate their difference, one could imagine two molecules, molecule A and molecule B, which in isolation smell like scent A and scent B respectively. Elemental odour perception would allow a human to smell at least one of, or both, scent A and scent B, while configural odour perception would allow a human to smell a scent that is neither scent A nor scent B.

A real example from Coureaud et al.’s paper describes an experiment wherein molecule A smells like strawberry and molecule B smells like caramel. An A/B ratio of 30:70 was perceived as more pineapple-like, showcasing the brain using configural processing at that ratio; however, at a slight alteration of said ratio, there was a decrease in the perception of pineapple. On the same note, an experiment highlighted in the Romagny et al. paper describes six different molecules that, when mixed together, present a red cordial smell not present in any of its components. However, at a ratio of 16.7% of each component, although the red cordial scent was still present, it was described as less typical of the red cordial scent. The authors hypothesize that an overshadowing effect had occurred within the mixture, whereby one component became perceptually dominant [36]. This effect is consistent with findings where altering component proportions leads to a perceptual shift toward one of the mixture’s individual components, which is more in line with elemental perception. The results of the experiments mentioned in these two papers could imply that the brain activates either configural or elemental processing depending on the concentration, or it could also mean that while the perception mode itself may not change, the typicality of a scent does.

Furthermore, Romagny et al. suggest that a human’s ability to perceive different odors in a single mixture has a ceiling of 3–4, wherein exceeding this might cause the brain to default to configural perception due to the capacity limitations of the brain [36]. Moreover, their paper also finds that different age groups perceive odors differently, with children leaning more toward elemental processing and adults toward configural processing, with teenagers falling in the middle; however, this developmental shift applies only to simple mixtures such as AB.

All of this suggests that a multi-molecule odor prediction model should be able to learn how molecules interact, since emergent scents cannot always be described by the individual components alone. This motivates the Set Transformer used in this study.

2.2.3 Perceptual Linearity vs. Receptor-Level Non-Linearity

While the papers mentioned in sections 2.2.1 and 2.2.2 assert the non-linear behavior of odorants when smelled, this subsection will introduce a paper that could seem to challenge this idea, arguing that the resulting scent from the combination of molecules, despite its perceived “configural” quality at the surface level, is still inherently mathematically predictable and therefore linear.

To prove this, Pellegrino et al. [32] built a quantitative linear model that takes in the unweighted average of an odorant’s descriptors, which are made up of a specialized lexicon of 51 specific but

broad olfactory labels, scored using a rate-all-that-apply system across 432 mixtures and 144 components, as opposed to Sisson’s binary presence/absence labels. Using Pearson correlation on the prediction of overall mixture quality, they were able to yield 0.71, which, when compared against the human benchmark, is not statistically different, proving that a linear model does work for this kind of prediction problem. Furthermore, they also tested famous yet unconventional odorant combinations, such as combining “tomato” and “cucumber” scents together to produce a “cooked fish” smell, yielding a relatively high score of 0.84. This is significant because, even though this kind of relationship would typically be used to describe nonlinearity, their linear model was still able to predict it.

Pellegrino et al.[32] explain this incongruence by claiming that what is causing this perception of non-linearity in odorants is the fact that they are described using simple words only, when in reality, the full olfactory characterization of a molecule holds much more dimensions than that. Taking the previous tomato and cucumber example, taking it at face value makes it seem like there is some nonlinearity when a cooked fish smell seems to develop out of nowhere. In reality, “tomato-smelling” and “cucumber-smelling” are not able to fully capture the complexity of what the scent profile of their respective molecules is really like. Each molecule also contained fish-like elements, but simple words are not enough to describe such occurrences, thus introducing the illusion of nonlinearity. These “hidden” scents exist and have computable values, and since mixing is linear, the hidden fishiness of both molecules was able to combine and thus seem to emerge from nowhere.

However, it is important to note that, despite these discrepancies in how odors are framed, such as whether an odor is configural or elemental, what Pellegrino et al. is trying to argue is that, at the base level, it is all the same when trying to predict the resulting smell, by including in the calculation the “hidden” semantic scent labels a molecule has. A better way to view the different angles that tackle linearity is that both works simply answer the problem differently: Pellegrino et al. uses RATA scores in an unweighted linear average calculation to predict a mixture’s overall quality, while Sisson et al. worked with binary descriptions and used GNNs to produce multi-label descriptors, which is more in line with what this thesis aims to achieve [32, 40]. Pellegrino et al. does acknowledge that, at the receptor level, there are nonlinear interactions that happen, which will be discussed further in a later section.

In any case, in simpler terms, odors could be considered linear when viewed through a continuous scale, while non-linear when viewed through a binary lens. Our thesis specifically takes the angle

of the latter to be able to produce multi-label predictions for odors, similar to Sisson et al.’s [2025] work, therefore justifying our choice of applying a Set Transformer for our prediction task.

2.3 Machine Learning on Sets and Chemical Mixtures

Having established the baseline information on computational olfaction and the behavior and psychophysics of odorant molecules when blended together, this next section discusses the machine learning perspective on the problem, specifically the mathematical foundation underlying the treatment of resulting scent compounds as sets and the relevant prior work on the topic.

2.3.1 Permutation Invariance on Sets

A combination of two or more molecules could be viewed as a mathematical set. An accurate depiction of a fragrance blend of these molecules requires that they be presented in such a way that their positioning in that list is arbitrary, not sequential [40, 46, 55]. For instance, assume the combination of molecule A and molecule B produces scent C. It should be of no relevance whether it is molecule A + B or molecule B + A; it should always produce scent C. For that reason, it is important for any mathematical model used to predict the resultant smell from the combination of different molecules to follow this principle, which is called permutation invariance.

Zaheer et al. [54] established the math of permutation invariance and introduced Deep Sets, a deep learning framework that uses a neural network designed to process datasets in a non-sequential manner for unordered data, which serves a different function from architectures that must process data in order (e.g., molecular compounds vs. protein sequences). This is referenced as the first of two properties they introduced in their study describing permutation invariance, positing that for a function acting upon a set, it must be indifferent to the order in which each element appears, mathematically formulated as $f : 2^X \rightarrow Y$ for any permutation, i.e., for any permutation π :

$$f(\{x_1, \dots, x_M\}) = f(\{x_{\pi(1)}, \dots, x_{\pi(M)}\}).$$

Their second theorem explains how to make the first property possible, stating that the function must be able to individually transform each element ϕ , the results of which are aggregated and then transformed again to ρ , mathematically formulated as decomposable to:

$$\rho \left(\sum_{x \in X} \phi(x) \right),$$

which guarantees the permutation invariance of a set [54].

However, as can be seen in the second theorem, the transformation of each element is done before pooling and is processed in isolation. Therefore, the information about how individual elements may interact with each other is not captured and gets lost, which is pointed out by Lee et al [26]. This is a critical issue in the context of fragrance blends specifically because of the non-linear behavior that odor molecules exhibit. The non-linear behaviors like emergence and masking are not captured if the molecules are treated as having no bearing on each other, which is false.

To bridge this gap, Lee et al. propose the use of a Set Transformer, where instead of processing the elements of a set individually, they apply self-attention mechanisms they call SAB (self-attention block) in the processing of the elements, wherein the different elements are able to learn from each other, such that the extracted features to be used for pooling already contain useful information about the elements’ relationship with each other. Moreover, they also developed PMA, or Pooling by Multihead Attention, which replaces the second step in the Deep Sets architecture wherein, instead of fixed sum or mean pooling, PMA also employs self-attention mechanisms in the pooling itself, such that it is able to pick out which of the elements bear more weight for the task at hand. Both of these innovations directly apply to our thesis problem. With the use of SAB, complexities of the interaction between odorant molecules are captured, and made meaningful with the use of PMA.

2.3.2 Precedents in Mixture Modeling

Zaheer et al. [54] provided the groundwork for permutation invariance in sets, while Lee et al. [26] extended it with self-attention, a general set method that later studies applied to molecular mixtures. Plenty of studies have made use of their work in different realms of chemistry.

For example, Qin et al. [33] built the SolvGNN architecture, which aimed to predict the activity coefficient between binary and ternary chemical mixtures by capturing their intermolecular interactions at two levels. First is the atomic-level (local) graph convolutions, which happen in the individual molecules, where features are extracted from each component atom, including the edges that connect them. This structural information is propagated and embedded in preparation for the next level, which is the molecular-level (global) message-passing graph, which operates at a broader scale as the nodes are now the molecules themselves, with the edges representing molecule-to-molecule interactions, such as hydrogen bonding. In this complete graph, message-passing algorithms are employed, using a Gated Recurrent Unit (GRU) to update each node with each pass, to capture

mixture compositions and non-ideal fluid behaviors, with Qin et al. [33] arguing that these dynamic measures are essential to explicitly model the real physical intermolecular forces in mixtures that naive pooling fails to do.

Similarly, Sanchez Medina et al. [38] introduce the Gibbs–Helmholtz Graph Neural Network (GH-GNN), which also predicts activity coefficients of chemical mixtures, but at different temperatures. What differentiates GH-GNN’s architecture from SolvGNN’s is that, instead of having two separate levels for GNN processing, GH-GNN has a single shared GNN, wherein the nodes and edges already contain both local and global vector features in a pair, which get updated over multiple layers of message passing. Afterwards, a global mean pooling operation is done to compress the data from each graph to create a vector embedding, concatenated with the updated global feature to create the final molecular fingerprint, which will then be transferred to another graph called the mixture graph, which maps out the hydrogen-bonding interaction between the solvent and solute molecules. Sanchez Medina et al. [38] defend this design by explicitly building the temperature-dependency relationship into the model’s structure through a Gibbs–Helmholtz-derived thermodynamic expression, rather than relying on a generic GNN to infer it implicitly from data alone. This closely mirrors, and inspired, our thesis’s approach of doing the same shared single-molecule encoder and then a separate mixture-level interaction step, but with odorant pairs instead of solvent and solute.

Lastly, Tom et al. [46] proposed POMMix, a more relevant precedent for our topic, as it is in the field of olfactory perception, aiming to predict the similarity of the scent of two mixtures. Theirs employs a 3-layer architecture, with the first making use of the previously discussed POM, where a GNN processes the atomic bond traits of the molecules along with the scent labels to create a multi-dimensional vector that describes the scent of the molecule. The second layer employs a module they call CHEMIX, which uses a self-attention mechanism across the graph from the first layer, doing set-pooling to produce a permutation-invariant mixture embedding that is able to tell which components contribute to the nonlinear behaviors that odorants exhibit. The last layer of their architecture is the cosine predictive head, which evaluates the similarity of mixtures according to their scent by calculating the scaled cosine distance between their respective embeddings. Their ablation study compared several molecular embedding types including handcrafted RDKit descriptors and pretrained language-model embeddings, combined with both a classical model (XGBoost) and their attention-based CHEMIX module, finding that embeddings derived from POM consistently outperformed the alternatives, reinforcing that structure-aware, attention-based representations capture

mixture behavior more effectively than simpler alternatives.

All three studies tackle three different prediction problems, yet use the same underlying concept of dynamic self-attention mechanisms, as they prove to perform better and better serve the problem of being able to model molecular interaction, as compared to basic aggregation and naive pooling/concatenation, going on to show the importance of using this framework for this paper, which aims to capture intermolecular interactions between odor pairs.

2.4 Biological Grounding of Olfactory Interactions

A scent is smelled when it travels into the air and to our noses, and inside our noses are about 400 different olfactory receptors that get these scents and deliver them to the brain [8, 12, 20, 24]. Building on the computational and machine learning frameworks used to predict chemical mixtures, this section explores the underlying biological mechanisms that govern these olfactory interactions.

In a nutshell, a receptor is like a road, and an odor molecule is like a vehicle. Different vehicles can go into many different specific roads, and many roads can accommodate various specific vehicles as well. When a car goes through the right road, it gets activated [8, 47]. However, there are also times when a vehicle enters a road but does not completely go through it, and then a different vehicle tries to go through the same road, but it cannot pass, because the other vehicle is currently blocking it. The same could happen with odor molecules, and this is called competitive antagonism. In more formal terms, it is when, instead of an activating ligand (agonist) binding on an active receptor or an orthosteric site to produce a biological response, what instead binds is a non-activating ligand (antagonist), which blocks or stops a response from happening [12, 53].

This phenomenon is explored and exemplified by multiple papers. For example, in the paper by Hu et al. [20], the molecule 2,3,5-trimethylpyrazine (TMP), which produces a roasted/nutty note, is supposed to activate receptor OR5K1, but instead is blocked by L-menthol (L-MT), which happens because of analogous binding conformations that prevent co-occupancy of the receptor. A similar thing happens in Fukutani et al. [12], where OR2T-family receptors, which were supposed to be activated by agonist malodors methanethiol (CH_3SH) and hydrogen sulfide (H_2S), instead get blocked by the antagonist β -ionone (a woody/floral ketone). Furthermore, they conducted a nostril experiment wherein: (a) both the antagonist and agonist were inhaled through a single nostril only, versus (b) the antagonist and agonist were inhaled separately through different nostrils. Strikingly,

in test (a), the perceived intensity of the agonist malodor was significantly decreased, compared to test (b), where no suppression occurred, which suggests that suppression in smelling odors happens at the receptor level, as opposed to the brain, and supporting the notion that this is because the two molecules were physically competing for a place at the receptor, rather than signals competing in the brain. This is also supported by another study, wherein the OR51E1 receptor was bound to by the antagonist Florhydral, reducing humans’ perception of the sweaty malodor isovaleric acid (IVA) [21]. This phenomenon is further supported by the Molecule to Olfactory Receptor (M2OR) database, which curates and compiles over 75,000 bioassay experiments and over 51,000 distinct OR-molecule pairs, detailing responsive agonists, non-responsive pairs, and other metadata that aid in the systematic exploration of the combinatorial nature of odorants in olfactory perception [24].

Taken together, these studies show that suppression in odor perception is a physical phenomenon that occurs when molecules compete over the same receptor. What this means for our study is that, just as a receptor’s response to one molecule can be blocked or weakened by the presence of a competing molecule at the same binding site, our Level 2 Set Transformer’s self-attention mechanism allows one molecule’s learned representation to influence how much weight another molecule’s features carry in the final blend embedding. This means that the Set Transformer is not only well-suited for handling molecular sets, but could also be used to model the mechanism of olfactory receptors in the nose, adding a real biological grounding layer to our architecture.

2.5 Prior Hierarchical GNNs

With the biological foundations of this study established, this section will discuss prior hierarchical GNNs in adjacent fields, which mostly operate on single-molecule or atom-to-fragment conditions, and how our proposed hierarchical GNN differs, as it operates between molecules, showing a single-molecule-to-blend hierarchy, which is a distinction that reveals a gap this thesis addresses.

Two formative works in the use of hierarchical graph neural networks for molecular property prediction are Hierarchical Informative Graph Neural Networks (HiGNN) and Hierarchical Molecular Graph Neural Networks (HimGNN) [16, 56]. HiGNN uses an algorithm called BRICS (Breaking of Retrosynthetically Interesting Chemical Substructures), which breaks down molecules into 16 different fragment types to create a graph, which is then combined with its global, non-broken-down counterpart graph to predict the molecule’s property [56]. Similarly, HimGNN makes use

of two graphs, with the first being a global graph as well, composed of nodes as atoms and edges as chemical bonds, and a motif-based graph, where the nodes are motifs, which are the specific molecular substructure patterns that frequently occur, and the edges are the atoms that share them. Both of these hierarchical GNNs are undeniably sophisticated, but they remain single-molecule, atom-to-fragment, meaning their scope is limited to a single molecule. There is other similar work employing GNNs that are non-hierarchical, such as MMGX (Multiple Molecular Graph eXplainable discovery), which employs four different graph topologies for multi-graph representation learning, but is still single-molecule in scope [23]. In olfaction, the previously discussed POMMix remains a rare exception that uses a hierarchical GNN for more than a single molecule; SolvGNN and GH-GNN do so for chemical properties rather than scent. This thus highlights the need for an approach that innovatively uses hierarchical GNNs for processing molecules as compounds.

2.6 Out-of-Distribution Splitting Strategies

This section will be the last technical segment of this chapter, providing a review of the standard data-splitting strategy of scaffold splitting, and an examination of how its application falls short in sparse chemical spaces, which then motivates the need for more rigorous out-of-distribution (OOD) evaluation. This section then grounds and discusses the clustering-based alternative our proposed framework will employ.

A common method to partition data for training, testing, and validation in chemoinformatics and molecular machine learning is scaffold splitting [14, 42]. In this method, molecules are represented by graphs of their structure, and the core framework of these graphs is referred to as their “scaffold” (Bemis–Murcko scaffolds) [51]. Molecules that share the same scaffold are grouped together, and these groups are then distributed among the training, testing, and validation sets to ensure that the test set contains structurally different molecules from the training set, which is a basic form of out-of-distribution (OOD) evaluation, which as opposed to random splitting, simply distributes molecules randomly across training, validation, and test sets [11, 14, 40, 42].

The caveat with this method is that, when the dataset is sparse, it collapses toward random splitting. For instance, similarly structured molecules get fragmented across both the train and test splits, leading to severe data leakage and inflated test scores as scaffolds in standard benchmarks could be populated by only one or two molecules each, undermining the grouping effect the method

depends on [11]. Another problem demonstrated by scaffold splitting is that even molecules that are near-identical (as computed by Tanimoto similarity) [14, 42], even differing by as little as a single atom, can, under this method, end up in opposite splits, as observed by Guo et al. and Steshin et al. [14, 42], who argue that this problem causes models to “memorize” rather than generalize from the data. Because of these problems, scaffold splitting can misleadingly produce scores that are overly optimistic.

To overcome these challenges, Joeres et al. [22] introduced DataSAIL, wherein the data-splitting problem was viewed as a mathematical optimization task; using a heuristic based on clustering and integer linear programming, they aimed to balance minimizing between-group similarity while maximizing within-group similarity, all while maintaining balanced fold sizes and class distributions. On the other hand, Shimakawa et al. [39] propose doing topological pre-clustering by projecting 2048-bit ECFP4 fingerprints onto a 2D map using 2D Uniform Manifold Approximation and Projection (UMAP), then grouping them together using spectral k -means clustering. This was then validated using Leave-One-Cluster-Out Cross-Validation (LOCOCV), where clusters are left out in rotation for training and testing, such that what the model was trained on does not show up in what it is tested on, meaning the model actually learns, rather than simply memorizing the data it was trained on, which is important for OOD evaluation.

However, while these methods may work well for single-molecule data splitting, they would be insufficient for multi-molecular fragrance mixtures, because an individual molecule can be a component of varying other molecules in different blends, meaning a molecule seen during training could reappear as a component in a different, unseen test blend, leaking information about that molecule across the split, causing the performance metrics to inflate. Therefore, Sisson et al. [40] propose “graph carving”, using a meta-graph to describe the pairwise/multi-molecule blend dataset, where each node represents an individual molecular graph and each edge represents a tested blend combination. Edges connecting a molecule from one partition to a molecule in another are discarded to guarantee no component overlap between splits. In their original implementation, this partitioning was done by randomly assigning molecules to each side and repeating until every label had at least one training and one test example, since a true scaffold-based split was not compatible with their sparse label distribution.

Building on this, our proposed framework instead applies the graph-carving protocol on top of the UMAP + k -means clustering discussed above, rather than random or scaffold-based partition-

ing. By applying this combined methodology to our merged GoodScents–Leffingwell–Sisson dataset, clustering molecules via UMAP and k -means, then carving the resulting blend meta-graph along those cluster boundaries, our proposed framework is evaluated on a strictly out-of-distribution test set.

2.7 Sustainable Fragrance Design and Environmental Considerations

As the architectural and methodological considerations have been discussed, this section positions this study as motivated by real-world problems which include the growing regulatory and environmental pressures affecting the fragrance industry, and the role computational tools can play in addressing them.

The development of fragrances is an intricate, costly, and time-consuming process, relying heavily on expensive, empirical trial-and-error methods that can take up to three years and cost up to \$50,000 per kilogram for perfumes specifically [35]. Beyond these hurdles, safety and environmental regulations facing the fragrance and flavor industry demand sustainable ingredient substitution [18]. Stricter global safety mandates, such as the European Union’s REACH regulations and evolving International Fragrance Association (IFRA) Standards, have caused legacy aroma-chemicals to become obsolete due to issues in material biodegradability, skin sensitization, and environmental toxicity [30, 46]. In response, initiatives such as the Safe and Sustainable by Design (SSbD) framework under the EU Chemicals Strategy for Sustainability were implemented to promote chemical safety and circularity across the product lifecycle.

To address these challenges, researchers have turned to Computer-Aided Molecular Design (CAMD) and Quantitative Structure-Property Relationship (QSPR) frameworks. However, as it stands, the selection guides used in traditional green substitution frameworks are static and contain fewer than 200 solvents, as expanding these guides would require collecting more than 30 distinct physical properties per molecule—a process that demands laborious laboratory testing [10]. Through the use of GNNs, which can dynamically generate descriptors for odorant molecules without the need for a priori engineering, in-silico pre-screening for candidate compounds can expand the searchable chemical landscape and significantly reduce laboratory trial-and-error processes [4, 10].

However, a candidate substitute would only be qualified for commercial fragrance formulation

if it meets two essential conditions: physical compatibility and smell potency [10, 27]. Physical compatibility refers to how smoothly the substitute dissolves in the base it is mixed into, and is evaluated using Hansen Solubility Parameters, where the Relative Energy Difference (RED) must be less than 1 [10, 35]. Potency matching, on the other hand, refers to predicting Odor Detection Thresholds (ODT) and Odor Activity Values (OAV), such that the target scent strength is achieved while balancing the safety and cost-effectiveness of the substitute’s concentration [27, 35]. These two requirements then serve as the basis for our Sustainable Substitute-Finder, which uses nearest-neighbor retrieval in the Level 1 GAT embedding space to find non-toxic, eco-friendly alternatives that preserve the original scent profile.

2.8 Synthesis and Research Gap

2.8.1 Synthesis of Literature

Current models in computational olfaction can successfully map single-molecule structures to their appropriate sensory descriptors [25]. However, because these compounds are evaluated in isolation, nonlinear psychophysical interactions such as masking and emergence—which determine what the final resulting scent would smell like when multiple odorant molecules are mixed—are not taken into account [20, 36, 40].

From an architectural standpoint, permutation-invariant set processing must be used to predict the behavior of multi-molecule mixtures [54, 26]. Standard linear pooling (mean/sum) methods are incapable of handling non-additive interactions due to symmetry collapse [44]. Current hierarchical GNNs, on the other hand, only consider intra-molecular fragments and fail to capture intermolecular blend interactions [16, 56].

Furthermore, standard scaffold splits suffer from severe data leakage at the component level in mixture spaces [11]. As a result, continuous manifold clustering must be used in conjunction with meta-graph carving to guarantee leak-free evaluation [39, 40].

2.8.2 Statement of the Research Gap

There is a lack of research evaluating an explicit, permutation-invariant attention mechanism (Set Transformer) against naive linear vector pooling on a standardized, leakage-resistant, out-of-distribution

(OOD) benchmark for multi-label fragrance blend prediction.

2.8.3 Positioning of the Present Study

To close this gap, this study presents a Structure-Aware Hierarchical Graph Neural Network (SA-HGNN) that applies a weight-shared Level 1 Graph Attention Network (GAT) to encode information on a per-molecule basis, and makes use of an explicit Level 2 Set Transformer (SAB + PMA) to handle the interactions between molecules, thereby allowing the model to computationally capture intermolecular interactions.

This framework is evaluated on a leakage-resistant Morgan-UMAP k -means graph-carved benchmark to quantify the value of explicit set attention over naive pooling, while establishing a green ingredient substitution engine under UN SDG 12 (Responsible Consumption and Production).

Chapter 3

Theoretical and Conceptual Framework

3.1 Theoretical Framework

This section establishes the mathematical and theoretical foundations underpinning the SA-HGNN architecture across four key areas. It first introduces the graph-theoretic formulation used to represent individual aroma molecules as attributed graphs, preserving natural atomic geometry and bond connectivity. It then details the deep learning theory behind Graph Attention Networks (GAT) for learning localized chemical representations, followed by the principles of set invariance and Set Transformers for modeling inter-molecular blend dynamics. Finally, it analyzes the expressivity limits of naive linear readouts, mathematically explaining how symmetry collapse degrades mixture representations and motivating the need for explicit cross-molecule attention.

3.1.1 Graph-Theoretic Molecular Representation

Aroma molecules will be modeled as an undirected, attributed molecular graph, enabling them to bond and create a mathematical representation where atomic and bond features are systematically updated via message passing [52]. This would allow the Graph Attention Network (GAT) to extract structural scent profiles regardless of atom ordering. In cheminformatics, chemical compounds are frequently written as one-dimensional character strings, such as text-based Simplified Molecular

Input Line Entry System (SMILES) strings. However, these text-based strings suffer from arbitrary starting atoms and are syntax-sensitive, which breaks up the natural symmetry of the physical molecule [31, 49]. Representing molecular structure as graphs preserves its true physical structure without depending on atom order, perfectly matching how our Graph Attention Network processes atomic connections.

Mathematical Formalism

Each aroma molecule will be defined as an undirected, attributed graph $G = (V, E)$, where:

- V represents the finite set of $N = |V|$ nodes, corresponding to the non-hydrogen atoms in the molecule. Hydrogen will be treated as implicit properties embedded directly within the feature vectors of their parent heavy atoms, rather than independent nodes, which will reduce the graph complexity and computational overhead [4].
- E represents the set of $M = |E|$ undirected edges. Each edge $e_{ij} = (v_i, v_j) \in E$ represents a covalent bond between heavy atoms v_i and v_j . Since covalent bonds represent a mutual sharing of valence electrons between atomic nuclei, the adjacency is symmetric, meaning $e_{ij} = e_{ji}$. These bonds or links are naturally symmetric, meaning the connection between them is bidirectional ($e_{ij} = e_{ji}$) [4].

The topological structure of the molecule is represented by a symmetric binary adjacency matrix $A \in \{0, 1\}^{N \times N}$, where A_{ij} values 1 if a chemical bond exists between atom v_i and v_j , and A_{ij} values 0 otherwise [56]. To incorporate rich chemical information, each node and edge is associated with a multi-dimensional feature vector:

- Node Vector $X_i \in \mathbb{R}^{46}$: a 46-dimensional profile recording the atom’s identity, charge, chirality, and immediate chemical environment [41, 56].
- Edge Vector $E_{ij} \in \mathbb{R}^{10}$: a 10-dimensional profile capturing the bond’s type, resonance, and spatial arrangement [56].

Olfactory Domain Mapping

To ensure the GNN can distinguish subtle chemical nuances that affect human scent perception—such as the difference between enantiomers or the presence of specific functional groups—comprehensive

node and edge features are generated via the RDKit cheminformatics library [41, 6, 25]. The exact composition of the 46-dimensional node feature vector and the 10-dimensional edge feature vector is detailed in Tables 3.1 and 3.2 below.

Chemical Property	Dimensions	Mathematical / One-Hot Representation & Details
Atomic Identity	9	One-hot encoding of key organic and halogen elements in flavor/fragrance compounds: [C, N, O, S, F, Cl, Br, I, Other].
Formal Charge	1	Scalar representation of the localized net electrical charge on the atom.
Atomic Degree	6	One-hot representation of the number of connected covalent bonds to heavy atoms: [0, 1, 2, 3, 4, 5+].
Explicit Valence	7	One-hot encoding of the total valence electrons committed to bonding in the molecule: [0, 1, 2, 3, 4, 5, 6+].
Hybridization State	6	One-hot representation of the orbital configuration: [sp , sp^2 , sp^3 , sp^3d , sp^3d^2 , Other].
Aromaticity	1	Binary flag indicating if the atom is part of a resonance-stabilized aromatic ring structure.
Attached Hydrogens	5	One-hot encoding of the number of hydrogens bound to this heavy atom: [0, 1, 2, 3, 4+].
Ring Size Participation	6	Binary flags representing specific ring sizes to map cyclic functional groups: [3-ring, 4-ring, 5-ring, 6-ring, 7-ring, 8-ring].
Ring Membership	1	Binary flag indicating whether the atom is enclosed within any ring structure.
Chirality Tag	4	One-hot representation of tetrahedral stereocenter configuration: [Unspecified, Clockwise (CW), Counterclockwise (CCW), Other].

Table 3.1: Specification of the 46-Dimensional Atom (Node) Feature Vector (X_i).

Chemical Property	Dimensions	Mathematical / One-Hot Representation & Details
Bond Type	4	One-hot representation of the bond's electronic sharing structure: [Single, Double, Triple, Aromatic].
Conjugation	1	Binary flag indicating if the bond belongs to a continuous, overlapping pi-electron system.
Ring Membership	1	Binary flag indicating if the bond is closed within a ring structure.
Stereochemical Config	4	One-hot representation of spatial configuration around double bonds: [None, Stereo-Z, Stereo-E, Other].

Table 3.2: Specification of the 10-Dimensional Bond (Edge) Feature Vector (E_{ij}).

Analytical and Methodological Justification

Aroma molecules modeled as graph structures instead of physical descriptors or using the text-based SMILES strings provide key advantages:

- **Permutation Invariance:** The model sees the connected web of atoms not as a sequential list but as a whole, ensuring it predicts scents no matter how the atoms are arranged [31].
- **Localized Feature Preservation:** Specific clusters of atoms, called “odorophores,” are the physical parts that plug into human nose receptors. The model captures every detail for every individual atom, allowing the network to preserve these local electronic environments and identify critical receptor-binding sites before aggregating them into global molecular embeddings [52, 25].
- **Structural Constraint Integration:** This enables the network to tell apart molecules with the exact same atomic ingredients but completely different 3D shapes and scents (such as isomers with identical atom counts but different bond orientations) [56].

3.1.2 Deep Learning Theory: Graph Attention Networks (GAT)

The model utilizes a Graph Attention Network (GAT) to map chemical structures to continuous vector representations. In a molecular graph, atoms act as nodes, while the chemical bonds connecting them act as edges. As the primary encoder, GAT helps the model to decide which neighboring atoms are most important to pay attention to when predicting a scent. Unlike older, traditional graph networks that statically average all atomic connections equally, GAT implements an anisotropic attention mechanism. This dynamic scoring system allows the network to place heavier weight on highly reactive regions—such as a polar oxygen group or a specific carbon ring—closely simulating the way physical human olfactory receptors target specific chemical features, which is called “odorophores”, rather than binding to a molecule indiscriminately [52, 25].

The GAT encoder represents molecules as attributed graphs, where atoms serve as nodes and bonds serve as edges. Let $\mathbf{h}_i$ denote the initial 46-dimensional feature vector of atomic node v_i . A shared parameter weight matrix $\mathbf{W}$ (of size $F' \times 46$, where $F' = 256$) first applies a linear transformation to project features into a highly expressive latent subspace. To determine the unnormalized attention coefficient $\tilde{e}_{ij}$ between a central node v_i and an adjacent neighbor v_j , a parameterized

single-layer feedforward neural network $\mathbf{a}$ evaluates their concatenated representations [52, 56]:

$$\tilde{e}_{ij} = \text{LeakyReLU}\left(\mathbf{a}^T [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j]\right). \quad (3.1)$$

In this formulation, $\mathbf{W}\mathbf{h}_i$ and $\mathbf{W}\mathbf{h}_j$ represent the projected features of the respective nodes, and the operator $\parallel$ denotes concatenation. The network then calculates how much attention one atom pays to an adjacent neighbor. This score is passed through a LeakyReLU function with a negative slope parameter of 0.2. This prevents inactive gradients across weakly interacting atoms by adding non-linearity to it [52].

A carbon atom can form up to four bonds, while a hydrogen atom forms only one. Atoms possess varying chemical valences which make them incomparable. To fix such thing, raw scores must be normalized across the immediate neighborhood $\mathcal{N}(i)$ of node v_i using a softmax activation function [52, 4]:

$$\alpha_{ij} = \text{Softmax}_j(\tilde{e}_{ij}) = \frac{\exp(\tilde{e}_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(\tilde{e}_{ik})}. \quad (3.2)$$

The normalized weights (α_{ij}) then scale the projected neighbor features during the localized aggregation. A non-linear activation function σ specifically, the Exponential Linear Unit ELU, is then applied to generate the updated representation for node v_i [52]:

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W}\mathbf{h}_j^{(l)} \right). \quad (3.3)$$

Finally, to stabilize the learning of the model, the GAT encoder implements $K = 4$ independent attention heads in parallel. In the hidden layers, outputs from different heads are combined to create a more detailed representation, and in the final readout layer, they are averaged to generate a consistent 256-dimensional molecular embedding [52]. To maintain adherence to spatial bond rules during the message-passing phase, the unnormalized attention approach is enhanced by directly including the 10-dimensional chemical bond feature vector $\mathbf{e}_{ij}$ directly:

$$\tilde{e}_{ij} = \text{LeakyReLU}\left(\mathbf{a}^T [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j \parallel \mathbf{W}_e \mathbf{e}_{ij}]\right). \quad (3.4)$$

This allows the network to distinguish subtle stereochemical features that are crucial for predicting the emergent scent profile of fragrance blends [52, 56].

3.1.3 Set Invariance and Set Transformers

Mixing two fragrance ingredients regardless of their order, the resulting scent will smell exactly the same. Mathematically, mixing Molecule A and Molecule B will have the exact same scent as mixing Molecule B and Molecule A. The blend follows the commutative property, which means that it behaves as an unordered set where the order of input does not change the output. To ensure that the blend prediction framework of this study follows the biological and chemical realism, it must be permutation-invariant [54, 40].

The Deep Sets Formulation

Deep Sets was established to solve this order-independence problem. It was mathematically proved that this framework guarantees permutation invariance by processing each set element through an identical neural network transformation before aggregating their representations using a symmetric function, such as summation and averaging [54]. Formally, a function $f(X)$ operating on a set of molecular embeddings X is defined as:

$$f(X) = \rho \left(\sum_{x \in X} \phi(x) \right). \quad (3.5)$$

The function ϕ represents an encoder network that projects each individual molecular embedding x to an intermediate latent space, while the sum operator aggregates these individual vectors [54]. This function is the level 1 Graph Attention Network (GAT) encoder in this study, turning each separate molecule into a 256-dimensional vector. This formulation guarantees order-invariance and is commutative because it treats each molecule of the same dimension. However, the translator ϕ evaluates Molecules A and B in isolation. Deep Sets assumes that these molecules do not affect each other. Real-world olfactory mixtures are complex and exhibit psychophysical phenomena such as scent masking and sensory emergence. The model is unable to capture the non-linear chemical interactions [40, 44].

Set Transformers and Attention-Based Pooling

The complex interaction between the molecules happens at the receptor level, which GAT alone cannot capture. To model these blending effects while maintaining permutation invariance, the model adapts the Set Transformer, which will allow molecular representations to interact dynamically

[26]. The Set Transformer employs two core blocks. The Set Attention Block (SAB) first allows molecule embeddings in a set to cross-attend to one another, enabling each molecule’s representation to be conditioned on its blend partner before pooling. This cross-molecule interaction captures psychophysical phenomena such as masking and emergence that a single encoder cannot model in isolation.

The second block is Pooling by Multihead Attention (PMA):

$$\text{PMA}_k(X) = \text{MAB}(S, \text{rFF}(X)). \quad (3.6)$$

This process begins by taking the molecular embeddings $\mathbf{X}$ and passing them through a row-wise feedforward network that prepares each molecule’s data for the final pooling calculations. The PMA block introduces a set of k learnable seed vectors, denoted as (S) . These probes are refined during training to actively extract the dominant features of the scent blend. The Multihead Attention Block (MAB) lets these seed vectors (S) attend to the polished molecule embeddings $(\text{rFF}(X))$, compressing the variable-size set into a fixed-size blend representation. This attention-based pooling captures the receptor-level interactions, providing a biologically and chemically realistic foundation for scent blend prediction [40, 26].

3.1.4 Readout Expressivity Limits and Symmetry Collapse

The chemical embeddings must be merged into a single representation to capture the exact blend of the molecules. Standard Graph Neural Network (GNN) designs are usually simple when merging vectors. Standard GNNs typically use linear permutation-invariant pooling, such as global sum or mean pooling [54]. Developments in representation theory prove that these impose a severe, encoder-independent expressivity barrier known as symmetry collapse [44]. The mathematical bottleneck limits the ability of standard GNN models from detecting psychophysical phenomena and non-linear chemical relationships, such as masking and emergence [40]. This argument concerns the mixture-level readout, where two molecule embeddings are combined; the within-molecule mean-pooling readout in Level 1 follows common practice in molecular GNNs and is not the focus of this study.

The Mathematics of Symmetry Collapse

Consider a molecular graph G whose set of node embeddings M is in a d -dimensional representation space $\mathcal{U}$. The structural symmetries of the graph are expressed by its automorphism group $\mathcal{H}$. It

consists of a list of all valid permutation node indices, whether flipped, rotated, or swapped; the molecules don’t change the way they are. When we apply a linear, $\mathcal{H}$ -invariant readout function f to compile these embeddings, the function is mathematically forced to factor through the group-averaging Reynolds operator (p_{avg}), defined as:

$$p_{\text{avg}} = \frac{1}{|\mathcal{H}|} \sum_{h \in \mathcal{H}} \rho(h). \quad (3.7)$$

The Reynolds operator projects the entire representation space $\mathcal{U}$ onto its fixed, trivial subspace $\mathcal{U}_1$ [44]. Consequently, any linear $\mathcal{H}$ -invariant map can be decomposed into the composition of a downstream linear map and the group-averaging projection:

$$f = f' \circ p_{\text{avg}}. \quad (3.8)$$

This factorization proves that a linear pooling readouts can only see what survives in $\mathcal{U}_1$. It completely projects away and discards the non-trivial, symmetry-aware components ($\mathcal{U}_2 \oplus \dots \oplus \mathcal{U}_m$) that capture node-to-node variation [44]. Even when the GAT encoder produces node embeddings that are genuinely sensitive to structural differences between molecules, none of that survives a linear readout at the mixture level. Molecular combinations with the same sum or the same average will smell and look alike after the structural information of the molecules is crushed, making it harder for standard GNNs to distinguish molecular combinations [44, 40].

3.2 Conceptual Framework

The conceptual framework of this study operationalizes the theoretical formulations into a systematic computational pipeline structured around the Input-Process-Output (IPO) model. Figure 3.1 outlines the structural interaction between input data representations, the two-level hierarchical deep learning architecture, and the downstream analytical and practical deliverables.

3.2.1 Input Phase

The computational data pipeline starts with the acquisition, curation, and gathering of data from olfactory databases. The raw SMILES strings are extracted from the GoodScents and Leffingwell repositories accessed via the Pyfrume Library in Python [6]. These raw chemical blueprints are then paired with expertly annotated odor tags to establish baseline inputs required from the pre-training

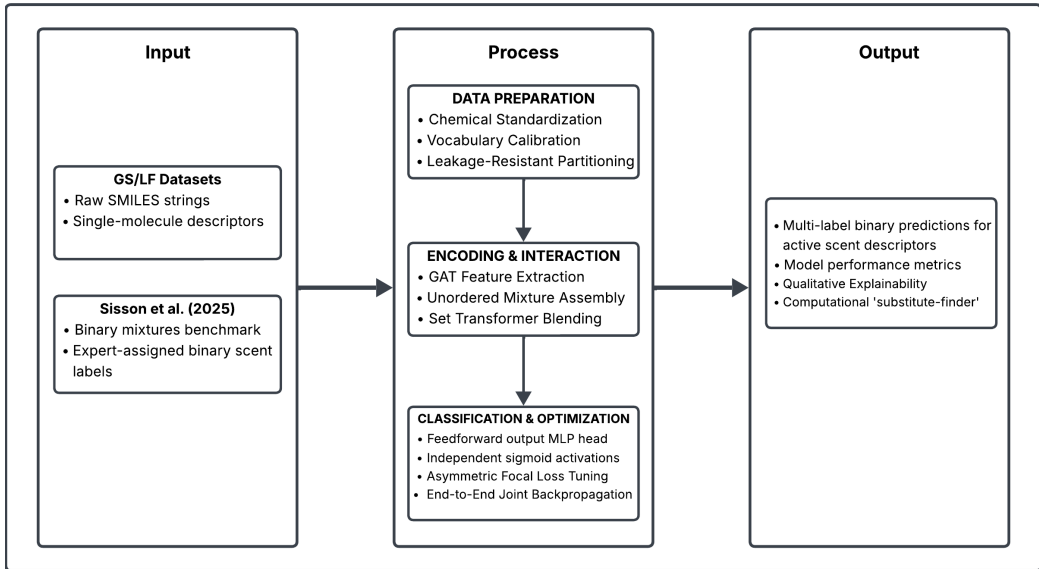


Figure 3.1: Input-Process-Output (IPO) Model.

phases. The binary mixture profiles are retrieved from the Sisson et al. (2025) benchmark dataset [40]. The retrieved benchmark dataset provides standardized chemical pair identifiers (CIDs) and multi-label scent descriptors that will help the network to learn and predict the combined scent of two scent blends.

3.2.2 Process Phase 1: Data Preparation and Graph Carving

To prepare the raw chemical input for model ingestion, it will first pass through a standardization pipeline. This stage eliminates non-target chemical debris to prevent evaluation bias. Salts and counterions are stripped using RDKit’s SaltRemover module, isolating the primary molecular structure. Compounds containing elements outside C, H, O, N, S, F, Cl, Br, and I are dropped, since these fall outside the range typical of aroma-chemicals [31]. The molecular weight should be between 20 and 600 g/mol. Below this range is considered irrelevant, while above the range is considered too heavy to be perceived as a scent at all [25]. The surviving SMILES strings are canonicalized using RDKit. The scent descriptors text from the source databases is standardized by merging redundant terms and eliminating descriptors with low frequency. This narrows down the label set to the core surviving scent classes determined by the support threshold gate across partitions [8]. To test

how well the network handles unfamiliar compounds, molecules are grouped into structural families by UMAP and K-Means on 2048-bit Morgan fingerprints [14]. The Graph-Carving algorithm then divides the mixture meta-graph along family clusters. This guarantees that validation and testing pairs do not contain any molecule encountered during training [11].

3.2.3 Process Phase 2: Encoding and Interaction

The second process phase converts the flat, two-dimensional 2D molecular graphs into continuous high-dimensional vector spaces and models non-linear interactions between paired compounds. In the Level 1 Graph Attention Network (GAT) primary encoder, individual atoms and bonds are processed as graph nodes and edges [52]. Since the Level 1 GAT shares its weights, it encodes both molecules under the same chemical rules, which then guarantees symmetry [38]. After encoding and updating atomic neighborhoods, global mean pooling condenses these vector spaces into a single 256-dimensional embedding for every compound [38, 52].

3.2.4 Process Phase 3: Classification and Optimization

Once the Level 2 yields the 256-dimensional blend vector spaces, it will then be translated into multi-label scent predictions using a three-layer neural network. This Multi-Layer Perceptron (MLP) head contracts sequentially from 256 to 128 and ends with L individual outputs to match the target scent classes [40]. Because scent profiles are inherently multi-label—where a mixture can simultaneously exhibit multiple scent characteristics—the classification layer utilizes independent sigmoid activation functions [37]. These activations compute individual probability scores for the joint presence or absence of each unique target scent descriptor. The system is trained end-to-end for network optimization. This means feedback from prediction errors flows backward through all layers simultaneously, fine-tuning both the Level 1 molecular encoders and the Level 2 mixture blocks guided with Asymmetric Focal Loss [17]. Olfactory datasets are highly sparse, meaning a single scent profile contains only a few active descriptors alongside dozens of inactive ones (zeros). Under standard training rules, the model would quickly learn that it can achieve over 97% accuracy simply by predicting “no” for every label. The asymmetric loss solves this class imbalance by ignoring these obvious negatives and forcing the network’s mathematical focus onto the rare, active scent notes that are difficult to predict [17].

3.2.5 Output Phase

The final phase outputs a multi-label binary vector of predictions across all L target scent descriptors. A validation module evaluates the model’s performance on these predictions under specified conditions. It follows and calculates metrics such as macro- and micro-averaged F1-scores, Hamming loss, and the Precision-Recall Area Under the Curve (PR-AUC) [3, 45, 48]. These metrics measure the model’s performance, ensuring that it generalizes well to new chemical structures. A mapping sequence extracts the weights generated by the Set Transformer for the researchers to study how the model is making decisions, shedding light on how the components interact, showing complex psychophysical behaviors, especially masking and emergence [20]. Finally, these grounded representations are fed into a sensory map to act as a sustainable substitute-finder. It scans designated molecular databases to discover environmentally friendly and safe chemical alternatives that are predicted to smell similar to hazardous ingredients [30, 18].

3.3 Technical Background

Mapping molecular structure onto multi-label olfactory predictions requires stitching together a few different layers of infrastructure: cheminformatics standards to make sure the molecules are valid, a graph learning framework to actually process them, and a loss function suited to how sparse the labels are [31]. This section walks through each piece—the parsing standards, the sparse graph representation, and the loss objective—and explains why each choice was necessary rather than incidental.

3.3.1 Cheminformatics Standard (RDKit): SMILES Parsing and Sanitization

This study utilizes the text-based Simplified Molecular Input Line Entry System (SMILES) strings used in molecular structures. SMILES strings are canonicalized and parsed into undirected, attributed molecular graphs. It uses the open-source RDKit library in Python [31]. Using these strings is convenient because they are compact and suitable for storage. However, this compactness comes at a cost: they are structurally sensitive, prone to syntax errors, and lack explicit physical geometry [49]. Before anything reaches the model, every string goes through RDKit’s sanitization

pipeline, which runs four checks in sequence:

- Kekulization checks that the alternating single and double bonds within conjugated or aromatic rings are represented in a chemically consistent manner, resolving ambiguous resonance structures.
- Aromaticity Perception evaluates the coordinate and topological structure of each atom to identify aromatic rings, dynamically reassigning aromatic bond attributes to relevant edges.
- Valence Checking checks every atom, specifically heavy atoms—carbon, nitrogen, oxygen, sulfur, and halogen to ensure it does not exceed its physical valency limits. This rejects syntactically corrupted SMILES strings or physically impossible hypervalent structures.
- Ring Perception, which identifies and indexes ring systems, information the later convolutions depend on.

By enforcing these rules, it ensures that what the network receives is something that is already known to be chemically valid, rather than forcing the neural network to infer the laws of organic valency implicitly from training data [4, 56]. The sanitization pipeline acts as an explicit structural filter for the model. This pipeline lets the Level 1 Graph Attention Network (GAT) operate exclusively on sparse, topologically accurate representations that map directly to real-world volatile compounds [25, 52].

3.3.2 Deep Learning Stack (PyTorch / PyTorch Geometric): Sparse Graph Convolutions

After sanitization, each molecule is formatted as an undirected graph $G = (V, E)$, where atoms represent nodes (V) and covalent bonds represent edges (E). A dense $N \times N$ adjacency matrix could represent this connectivity, but for molecules, where most atoms bond to only a few neighbors, that approach wastes memory, scaling at $\mathcal{O}(N^2)$ for structures that are inherently sparse. The architecture instead runs on PyTorch Geometric, which handles molecular graphs as sparse attributed objects rather than dense arrays [31, 49].

Topology is stored in Coordinate (COO) format: a sparse two-row index matrix C of size $2 \times 2E$, where the first row lists source node indices and the second lists receiver node indices [4]. Since chemical bonds are physical and undirected, every source node is also a receiver, which is why $2E$

entries are needed rather than E — each bond has to be represented in both directions for message-passing to work [4]. Node states are then updated by running sparse convolutions over this index matrix, a process that breaks down into three phases at layer l .

- **The Message Phase:** For each neighbor v_j connected to a central node v_i , the network builds a directed message from their hidden states plus the bond features e_{ij} connecting them [56]:

$$\mathbf{m}_{ij}^{(l)} = \text{Message}^{(l)}(\mathbf{h}_i^{(l)}, \mathbf{h}_j^{(l)}, \mathbf{e}_{ij}). \quad (3.9)$$

- **The Aggregation Phase:** All messages arriving at node v_i from its neighborhood $\mathcal{N}(i)$ get combined. Because graphs have no fixed ordering or coordinate system, this step has to be permutation-invariant — the result can't depend on what order the neighbors happen to be indexed in. Here, aggregation is guided by GAT attention coefficients α_{ij} , which weight chemically active neighbors more heavily than inert ones [52]:

$$\mathbf{M}_i^{(l)} = \sum_{j \in \mathcal{N}(i)} \alpha_{ij} \cdot \mathbf{W}_{\text{msg}}^{(l)} \cdot \mathbf{h}_j^{(l)}. \quad (3.10)$$

- **The Update Phase:** The aggregated message is combined with the node's own current state to produce its representation for the next layer, passed through an ELU activation to give the model room to capture nonlinear structure [4, 11, 52]:

$$\mathbf{h}_i^{(l+1)} = \text{ELU}(\mathbf{M}_i^{(l)} + \mathbf{W}_{\text{self}}^{(l)} \cdot \mathbf{h}_i^{(l)}). \quad (3.11)$$

3.3.3 Multi-Label Mathematical Formulation:

Independent Sigmoids and Sparsity Mitigation

Since fragrance smells are complex and rarely just smell one thing, a mixture can read both smells at once. To handle this, the final layer skips a mutually-exclusive softmax and instead maps the 256-dimensional blend embedding through an MLP to L independent sigmoid outputs, one per target scent descriptor [37, 40, 45, 48]. Each output is a probability $y_{\text{pred},k}$ bounded between 0 and 1:

$$y_{\text{pred},k} = \sigma(x_k) = \frac{1}{1 + e^{-x_k}}. \quad (3.12)$$

A standard multi-label setup would optimize this with plain Binary Cross-Entropy, summing error across all L target classes independently [37]:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{K} \sum_{k=1}^L \left[y_k \log(y_{\text{pred},k}) + (1 - y_k) \log(1 - y_{\text{pred},k}) \right]. \quad (3.13)$$

However, olfactory data is severely imbalanced. Label density across the master dataset is typically below 3% [37], meaning a typical molecule or blend has only a few active descriptors out of the candidate vocabulary, with the vast majority sitting at zero. Across the whole matrix, that puts zeros at over 97% [3, 8]. Under plain BCE, those easy negatives dominate the gradient — a model can hit over 97% accuracy just by predicting zero everywhere, and in doing so never learns the rare positive labels that are actually informative.

To resolve this, the pipeline uses an Asymmetric Focal Loss, which splits the focusing behavior for positives and negatives into separate exponents, γ_p and γ_n , and adds a class-frequency weight α_k [17]:

$$\mathcal{L}_{\text{FL}} = -\frac{1}{K} \sum_{k=1}^L \left[y_k \cdot \alpha_k \cdot (1 - y_{\text{pred},k})^{\gamma_p} \cdot \log(y_{\text{pred},k}) + (1 - y_k) \cdot (1 - \alpha_k) \cdot y_{\text{pred},k}^{\gamma_n} \cdot \log(1 - y_{\text{pred},k}) \right]. \quad (3.14)$$

Setting $\gamma_p = 0$ leaves gradients from the rare positive labels untouched — they pass through at full strength. Setting $\gamma_n = 2$ does the opposite job for negatives: it exponentially suppresses the loss contribution from easy, correctly-predicted zeros, which is what strips away their gradient dominance. Together, these two settings push the model’s attention toward the rare positive notes instead of letting it coast on negatives, which is what makes reliable multi-label prediction possible on data this sparse [17].

Chapter 4

Methodology

This chapter explains the methodology adopted for designing, implementing and evaluating the structure-aware hierarchical graph neural network (SA-HGNN) proposed in this research work. First, the chapter discusses the overall research methodology, highlighting the variables (both independent and dependent) studied and controlled variables in this experiment. After that, it discusses the data collection and harmonization methods, which include the integration of Goodscent, Leffingwell and Sisson et al.’s [40] data sets into a combined dataset. Next, the chapter discusses the technical methodology, including the model and architecture, explaining the architecture configuration of Level 1 GAT encoder, Level 2 Set Transformer and classification layer that generate multi-label scent predictions. Then, the chapter tackles the out-of-distribution splitting and graph carving methodology used for generating a leakage-resilient test set. Finally, it discusses the metrics and statistical significance tests, and the ablation study design, used for analyzing the contribution of Level 2 Set Transformer.

4.1 Research Design

This study operates through a Quantitative Experimental Framework, employing supervised learning techniques on a multi-label molecular graph classification task. In this study, the independent variable is the Level 2 aggregation method, with three levels: the Set Transformer (the proposed method), mean pooling, and a reproduced pairwise blend model (the baselines). These are evaluated using four dependent variables: Hamming loss, macro F1, micro F1, and PR-AUC. The controlled

variables are the Level 1 GAT encoder, the data split, the loss function, the training hyperparameters, and the random seeds. By doing so, the effect of the aggregation mechanism is isolated in order to ensure that any difference in performance observed could not be attributed to anything but the interaction modeling technique.

4.2 Data Acquisition and Harmonization Pipeline

This section outlines the general procedures for collecting the raw single-molecule data from Leffingwell and GoodScents and blend-level data from Sisson et al.’s. It details how these data are merged, cleaned, and standardized to create one harmonized source.

4.2.1 Dataset Consolidation

Data acquisition will focus on three main datasets: Goodscents, Leffingwell, and Sisson et al.’s [2025]. The first two, accessible through Pyrfume, serve as the single-molecule sources for Level 1 pretraining and as the lookup reference for the molecular structures referenced in Sisson’s dataset. Sisson et al.’s dataset, accessible through their GitHub repository, serves as the reference for molecule blends used in Level 2 fine-tuning and evaluation.

Each of the two single-molecule datasets contains three files, namely: `behavior.csv`, which contains the stimulus and descriptor pairs data; `molecules.csv`, containing CID, MolecularWeight, IsomericSMILES, IUPACName, name; and the `stimuli.csv` file, which stores data such as Stimulus, TGSC ID, CID, Concentration %, Solvent.

The Leffingwell and Goodscents data files will be merged into one master dictionary, joined by the IsomericSMILES key present in both their `molecules.csv` files, where every molecule has both its canonical structure and its full set of odor descriptors attached. Every IsomericSMILES string in both sources is canonicalized with RDKit so that the same molecule written in different ways is matched correctly. Blend components in Sisson et al.’s dataset, which are identified by PubChem CIDs, are linked to the master dictionary by canonicalizing their SMILES in the same way. This was the key chosen as opposed to CID, as some of the CIDs contained in the files were negative/non-standard and would require further standardization, while the SMILES would already be canonicalized either way in the preparation stage for converting them into graphs. Furthermore, data such as the Concentration % and Solvent from the `stimuli.csv` files are to be dropped, since

concentration is scoped out from this study.

Descriptors from behavior.csv, which are indexed only by Stimulus, will first be joined to stimuli.csv on the shared Stimulus field to obtain each entry’s CID, which is then used to join to molecules.csv, which completes the within-source link between a compound’s structure and its descriptors before the cross-source SMILES merge described above takes place.

4.2.2 Vocabulary Standardization

To be more precise, there are approximately 669 unique raw string labels from Goodscents and 114 from Leffingwell, with “unique” here meaning distinct as literal text, since spelling variants of the same underlying descriptor (e.g., anisic/anise) are not yet consolidated at this stage. Upon closer inspection, there is a normalization problem arising wherein sources contain duplicate spellings or variants of scent descriptors, such as anisic/anise, wine/winey, cherry/cherry maraschino cherry, and caramel/caramellic. Manually reviewing each of the scents would be arduous at this scale; therefore, this study will adopt Sisson et al.’s own open-source canonicalization dictionary, which standardizes approximately 400 raw descriptors down to 104.

As the merged data may contain scent descriptors outside this dictionary, a fuzzy string-matching or edit-distance comparison using rapidfuzz’s Levenshtein-based similarity scoring will be done across the residual, uncovered portion of the combined raw label set to identify likely duplicate descriptors. Once identified, they will be manually reviewed and resolved, deciding whether each is mapped to an existing note or declared a genuinely new descriptor, with the latter still subject to being dropped if it does not appear across at least 30 molecules. Further normalization checks will be done, such as whitespace trimming and punctuation stripping.

This standardization applies to the single-molecule (Level 1 pretraining) vocabulary, which is permitted to exceed 104 notes if genuinely new descriptors survive the residual fuzzy-matching process described above. This is distinct from the blend-level (Level 2) classification target, which remains fixed at whatever the resulting count will be when the same survival criterion Sisson et al. applied to their carving run that narrowed down their 104-label vocabulary up to 74 is applied. This paper’s split may yield a similar but not necessarily identical count.

4.3 Technical Modeling and Architecture

This section presents the technical architecture of SA-HGNN in the order data flows through it: first Level 1, the Graph Attention Network that encodes each molecule independently from its atomic and bond-level features; then Level 2, the Set Transformer that allows the two resulting molecule embeddings to interact; and finally how the combined blend embedding is converted into per-descriptor predictions, along with the loss function used to train it.

4.3.1 Level 1 GAT Featurization

Once the dataset is read, the merged GoodScents and Leffingwell molecules are first clustered and split using UMAP and K-Means (Section 4.4.2), and only the molecules in the training clusters enter the Level 1 GAT featurization phase for standalone pretraining, so molecules held out for validation and testing are never seen during pretraining. Each molecule is represented as a graph with 46-dimensional node features and 10-dimensional edge features (Section 3.1.1, Tables 3.1–3.2). This GAT has three layers and uses LeakyReLU to compute raw attention scores between connected atoms, and normalizes these scores with the softmax formulation in Section 3.1.2.

Once the GAT processes a molecule’s graph, via a mean-pooling readout, its resulting 256-dimensional molecule vector passes through a Multi-Layer Perceptron that maps it to one output per standardized single-molecule descriptor, producing independent sigmoid outputs; the exact number of descriptors is reported after vocabulary standardization (Section 4.2.2).

Afterwards, a Binary Cross-Entropy loss function will be employed for backpropagation. This will be used as opposed to Focal Loss because the latter is more geared toward preventing models from becoming lazy and predicting all-negative outputs, a problem that worsens after Level 2’s graph-carving step. Focal Loss will instead be reserved for Level 2, described in Section 4.3.3.

After pretraining is completed, this temporary output head is discarded, leaving only a GAT with its encoder weights trained. In this step, the model is trained with the single-molecule dataset until it is ready for Level 2, wherein it enters a two-phase protocol for learning blend mechanics.

4.3.2 Level 2 Set Transformer Configuration

With Level 1 delivering a GAT trained on single molecules, the same GAT is applied to the component molecules of a blend pair, once per molecule, producing a 256-dimensional embedding for each:

$\mathbf{z}_A, \mathbf{z}_B \in \mathbb{R}^{256}$ Using the same GAT for both molecules ensures they are embedded under identical rules regardless of order. At this point, the two embeddings form an unordered set:

$$S_{\text{blend}} = \{\mathbf{z}_A, \mathbf{z}_B\} \subset \mathbb{R}^{256} \quad (4.1)$$

and it is the Set Transformer’s subsequent processing, rather than the shared GAT alone that guarantees the blend prediction itself is permutation-invariant: $\{\mathbf{z}_A, \mathbf{z}_B\} = \{\mathbf{z}_B, \mathbf{z}_A\}$ must yield the same result.

This set is passed through the Self-Attention Block (SAB), computing standard attention using Queries (Q), Keys (K), and Values (V) derived from the set itself:

$$\text{SAB}(X) = \text{MultiHead}(X, X, X) \quad (4.2)$$

As in the attention mechanism introduced in Section 3.1.2, each element attends to every other element in the set by comparing its Query against the Keys of all elements (including itself) to compute attention weights, which are then used to combine their Values into an updated representation. Because the same set X serves as the source of Q, K, and V simultaneously, this is self-attention rather than attention across two distinct sets. Concretely, for the two-element blend set, molecule A’s Query is compared against both A’s and B’s Keys, producing attention weights that combine A’s and B’s Values into an updated embedding for A with the same process occurring for B.

SAB is configured with 2 blocks and 4 attention heads, where the two embeddings are no longer treated as individual molecules but as components of a blend. Here, molecules A and B "learn" from each other via message passing, using 4 attention heads in parallel per block, across 2 sequential blocks, with each pass updating both embeddings based on the other’s influence. Formally, this yields the updated embeddings $\mathbf{z}'_A$ and $\mathbf{z}'_B$, where each is now a function of both original embeddings rather than of its own molecule alone. By the end of this process, $\mathbf{z}'_A$ and $\mathbf{z}'_B$ are no longer independent, having been shaped by each other’s influence, designed to allow the model to represent the receptor-level masking and competition discussed in Chapter 2.

Afterwards, $\{\mathbf{z}'_A, \mathbf{z}'_B\}$ undergoes Pooling by Multihead Attention (PMA), compressing the set into a single 256-dimensional blend embedding $\mathbf{z}_{\text{blend}} \in \mathbb{R}^{256}$. This compression uses attention instead of simple averaging so that it is possible for one molecule’s contribution to be weighted more heavily than the others depending on the learned relationship between them (e.g., a dominant, overpowering scent molecule masking a weaker one).

4.3.3 Output Head & Asymmetric Focal Loss:

The final blend embedding $\mathbf{z}_{\text{blend}} \in \mathbb{R}^{256}$ is passed through a new Multi-Layer Perceptron that contracts from $256 \rightarrow 128 \rightarrow L$, where L is the number of descriptors that survive the leakage-resistant OOD split described in Section 4.4, with up to 74, following Sisson et al.’s own carving outcome, with the exact value reported once this study’s own split is executed.. The output layer applies independent sigmoid activations across all L nodes, $x \in [0, 1]$, since scent descriptors are multi-label: each descriptor must be treated as an independent binary decision (1 = present, 0 = absent), unlike softmax, which forces all outputs to sum to 1.0, which is a formulation consistent with the multi-label framework established in Section 3.3.3. The resulting predictions are optimized using the Asymmetric Focal Loss defined in Section 3.3.3, with $\gamma_p = 0$ and $\gamma_n = 2$, following Ameta et al. [3].

As such, the final pipeline proceeds as follows for fine-tuning: single-molecule data is featurized and converted into a graph for Level 1; the same GAT trained on single molecules is then reused on those same molecules as components of a blend pair, aggregated together via SAB and PMA, and passed through the L -node sigmoid output head; predictions are scored using Asymmetric Focal Loss, with the resulting gradients updating weights across the entire network, including the pretrained Level 1 GAT end-to-end.

4.4 Leakage-Resistant OOD Splitting and Graph Carving

This section describes how the dataset is partitioned into training, validation, and test sets such that the resulting evaluation genuinely measures generalization to unseen chemistry, directly addressing the second research question. It explains the reason for why conventional scaffold split is unsuitable for this dataset, then describes the chemical-fingerprint clustering approach used instead, the graph-carving procedure that removes any blend pair spanning two partitions, the descriptor survival filter applied after carving, and finally a sanity check comparing this split against a standard random split.

4.4.1 Scaffold Split Decay

As discussed in more detail in Section 2.6, scaffold splitting does not work well for out-of-distribution (OOD) evaluation in sparse chemical spaces, where scaffold groups contain too few constituent

compounds, causing it to collapse to just random splitting instead. This step verifies this claim mathematically as opposed to just relying on the aforementioned literature. To do so, the molecule-to-scaffold ratio (R), is defined as:

$$R = \frac{N_{\text{molecules}}}{N_{\text{scaffolds}}} \quad (4.3)$$

where $N_{\text{molecules}}$ is the total number of unique canonicalized compounds, and $N_{\text{scaffolds}}$ stands for how many distinct Bemis–Murcko frameworks were generated using RDKit. As R gets closer to 1.0, it means that almost every scaffold only contains one molecule, meaning that there is virtually no shared molecular framework across the compounds. R will be computed based on the merged Goodscents + Leffingwell master dictionary during data prep and will be benchmarked against Fooladi et al.’s own collapse threshold of $R < 2.0$. Our R falling below theirs would further justify using the clustering-based alternative (Morgan-UMAP K-Means graph carving) detailed in the next section, rather than scaffold splitting.

4.4.2 Morgan-UMAP K-Means Split

To represent each molecule, ECFP4 fingerprinting will be performed via RDKit for all unique molecules in the dictionary, generating 2048 binary flags with a radius of 2, capturing atomic environments up to two bonds away.

These 2048-bit ECFP4 fingerprint vectors will then be mapped into a continuous 2D coordinate space using Uniform Manifold Approximation and Projection (UMAP), making it easier to cluster, with chemically and structurally similar molecules sitting closer together and vice versa. K-Means clustering will then be applied to the 2D map with $k = 7$. This value of $k = 7$ follows Guo et al. [14], who found seven UMAP clusters to outperform alternative clustering methods on chemical molecule data.

The OOD test-cluster selection rule works as follows: among the 7 candidate clusters, only those satisfying the requirement that every retained descriptor has at least one training instance and one test instance after graph-carving are considered eligible. Among these eligible candidates, the cluster selected is whichever one retains the largest number of usable blend pairs, as opposed to selecting based on raw cluster size. This mirrors Sisson et al.’s own justification for their chosen split ratio, which similarly prioritized maximizing usable data and label coverage over a fixed proportion.

A second eligible cluster, the one retaining the next-largest number of usable blend pairs, is withheld as the validation cluster, and the remaining clusters form the training set in the selected

test and validation clusters are withheld from training, establishing the single-molecule boundary required for the Graph-Carving procedure described in Section 4.4.3.

4.4.3 The Graph-Carving Protocol

The Sisson et al. [40] pairwise blend dataset is treated as a meta-graph in the construction of the graph-carving procedure, wherein each node represents a unique canonical single molecule, and each edge represents a documented blend between two such molecules.

Carving is then performed by examining each edge: if the two molecules an edge connects do not lie within the same partition (training, validation, or test), that edge is discarded. Edges are retained only if both molecules of a blend pair belong to the same partition to ensure there is no component overlap between any two partitions, preventing leakage. The validation set will be used for early stopping and will be the basis for hyperparameter tuning and the decision threshold for the sigmoid outputs. The test set will be used only once, in the final evaluation.

This carving mechanism mirrors the approach Sisson et al. [40] applied to their own random 50:50 partition, which discarded approximately 83,000 pairs to yield 44,000 training and 40,000 testing pairs. Nevertheless, since this thesis will group its chemical space using the Morgan-UMAP K-means clustering method mentioned in section above, instead of a randomly split 50:50 set, the exact pair count and the label count that will survive the pipeline will only be noted after it is executed.

4.4.4 Survival Vocabulary Filter

With graph carving discarding a significant amount of data from the dataset, the frequencies at which descriptors appear are altered. There is a need for a filter strictly targeting the blend-level target matrix, ensuring that every descriptor in the final multi-label target vector is sufficiently represented in its positive appearances on both sides of the carved split, for meaningful training and evaluation. This is determined by:

$$d \in \mathcal{D}_{\text{eval}} \iff c_{\text{train}}(d) \geq x \quad \text{and} \quad c_{\text{test}}(d) \geq x \quad (4.4)$$

wherein $c_{\text{train}}(d)$ is the number of instances descriptor d is positive in the carved training partition $\mathcal{E}_{\text{train}}$, and $c_{\text{test}}(d)$ is the number of instances descriptor d is positive in the test partition. x , where x , set to $x = 5$, meaning each retained descriptor must be positive in at least 5 blends on both

the training and test side. x is the minimum support threshold, where x dictates the minimum number of positive blends required on each side. A descriptor failing this condition on either side $c_{\text{train}}(d) < x$ or $c_{\text{test}}(d) < x$ is dropped from the classification target matrix. This emulates the filtering logic Sisson et al. [40] applied to their own carving run, where their final set of 74 emerged from applying this same survival criterion to their standardized 104-label vocabulary.

4.4.5 Interpolative Baseline Sanity Check

This step addresses the question of how much performance the model loses specifically because of the OOD splitting done, versus using a naive random splitting technique. To do so, the same single-molecule and blend datasets are split three ways (training, validation, and test) at random instead of by UMAP cluster, while still applying the same meta-graph carving and survival filtering processes. Everything else also stays the same, such as the model architecture, training hyperparameters, loss functions, and evaluation metrics.

The OOD performance decay Δ_{OOD} is calculated using:

$$\Delta_{\text{OOD}} = \text{Metric}_{\text{random}} - \text{Metric}_{\text{OOD}} \quad (4.5)$$

wherein a large gap ($\Delta_{\text{OOD}} \gg 0$) means that random splitting does cause performance inflation because structurally similar molecules can fall on both sides of a random split. A small gap ($\Delta_{\text{OOD}} \approx 0$) indicates that the proposed GNN framework is able to generalize meaningfully even when confronted with structurally unfamiliar chemical families it has never encountered during training. For Hamming loss, where lower is better, the difference is computed as OOD score minus random-split score, so that a positive Δ still indicates the OOD result is worse. This step is purely secondary, and the UMAP-clustered OOD split remains the study’s primary reported evaluation benchmark.

4.5 Quantitative Metrics and Metric Invariance

This section defines the metrics used to evaluate model performance and explains why each is necessary given the severe label sparsity of the blend-prediction task. It first examines the limitations of precision, recall, and F1 under this sparsity and motivates the use of macro-averaged F1 as the primary metric, then introduces Hamming loss as a complementary metric and defines it formally.

4.5.1 Multi-Label Metric Invariances

Let TP (true positive) be a descriptor correctly predicted as present, FP (false positive) a descriptor incorrectly predicted as present, FN (false negative) a present descriptor the model missed, and TN (true negative) a descriptor correctly predicted as absent. Because only about 2 to 4 of the L descriptors are positive in a typical blend, an estimated 95% of all label entries are negative. Because of this, a model can look like its accurate simply by saying a label is “absent” every time.. Precision, recall and F_1 are computed only from TP, FP and FN [34], so they ignore the many easy true negatives. A model that predicts no descriptors has no true positives and therefore receives an F_1 of zero, which exposes this failure. For this reason, macro-averaged F_1 , which gives every descriptor equal weight regardless of how common it is [34], serves as the primary metric. Micro-averaged F_1 pools all predictions and summarizes overall performance, and PR-AUC measures how well the model ranks true descriptors above false ones across all decision thresholds.

4.5.2 Hamming Loss Integration

Hamming loss is also reported, as it is a standard multi-label metric [34] that counts every incorrect label decision over the full $N \times L$ label matrix. Because it also counts the many easy true negatives, a model that predicts no descriptors would still obtain a low Hamming loss of about 0.03. Hamming loss is therefore interpreted alongside the all-negative and label-frequency reference baselines rather than on its own. This is calculated by the formula:

$$\text{HL} = \frac{1}{N \cdot L} \sum_{i=1}^N \sum_{j=1}^L \mathbf{1}(y_{ij} \neq \hat{y}_{ij}) \quad (4.6)$$

Where N is the total number of evaluated blend samples, L is the number of surviving target descriptors, up to 74 per Sisson et al.’s own carving outcome (Section 4.3.3), with the exact value reported once this study’s own carving is executed $\hat{y}_{ij} \in \{0, 1\}$ is the model’s binary prediction given a decision threshold, and $\mathbf{1}(\cdot)$ is the indicator function (returning 1 if true, 0 if false). Smaller values indicate higher accuracy ($\text{HL} \in [0, 1]$, with $\text{HL} = 0$ representing a perfect match).

4.6 Statistical Significance Testing

This section addresses the risk that the metrics gathered from evaluation might reflect random noise, such as seeding or splitting luck, rather than real performance differences, using inferential statistical

testing. Recent work cautions against relying on naive paired t -tests for comparing machine learning models, since the t -test is strongly affected by outliers and is not valid when resampled test sets are used [34]. Instead, this study will employ bootstrap resampling, as recommended by recent work, since it shows both whether one model is better than another and by how much [5]. The Set Transformer and the baseline models will be scored on the same resampled test indices, so that any difference reflects the models themselves rather than the test samples drawn. To account for seeding luck, each model will also be trained with several random seeds.

To do so, the fixed, held-out test predictions of N blend instances are taken, and N predictions are sampled with replacement 10,000 times, wherein both the Set Transformer and the baseline model are evaluated on the same resampled indices to ensure a paired comparison. For each bootstrap iteration b , the paired metric difference $\Delta(b)$ is computed by:

$$\Delta(b) = \text{Metric}_{\text{SetTransformer}}(b) - \text{Metric}_{\text{Baseline}}(b) \quad (4.7)$$

yielding a distribution of 10,000 paired differences that characterizes the variability of the Set Transformer’s advantage over the baseline. This procedure is repeated independently for each metric in the 4-metric suite from Section 4.5.2.

The significance threshold will be set at $\alpha = 0.05$ and 95% Empirical Confidence Intervals (CIs), which show the plausible range of the true underlying performance difference, will be extracted from the 2.5th and 97.5th percentiles of the 10,000 bootstrap iterations. Here, a CI that does not cross zero indicates a statistically significant positive gain at $\alpha = 0.05$. Since statistical significance alone does not show whether a difference is practically meaningful, the size of the effect is reported as the mean difference together with its 95% CI, expressed in the units of each metric.

This significance testing framework will be deployed for the purpose of comparing the Level 2 Set Transformer against naive Set Readouts (Mean Pooling) across the 4-metric suite from Section 4.5.2.

4.7 Ablation Study Design

This section describes the ablation and comparison studies used to isolate the contribution of each architectural component, which directly addresses the first and third research questions. It begins by replacing the Level 1 GAT encoder with a simpler Graph Convolutional Network to test the value of

attention at the atom level, then compares the proposed Set Transformer against mean pooling, and a reproduced pairwise blend model to test the value of attention at the molecule level, and finally examines whether the resulting attention weights align with documented receptor-level biology.

4.7.1 Intra-Molecule Encoder (Level 1 GCN Ablation)

This section details the ablation study choices that will be done on the model, where individual architectural components are swapped out to isolate their contribution to performance.

Specifically, only a partial ablation study will be conducted at Level 1, wherein the Level 1 GAT encoder is replaced with a Graph Convolutional Network (GCN). This is tested in combination with the winning Level 2 configuration discovered in the following subsection 4.7.2.

Level 1 undergoing only a partial ablation is a deliberate choice, as the encoder it uses is only secondary to the main question this thesis is asking, which is about how molecules in a blend interact with each other, being addressed by Level 2, not Level 1. Doing a full factorial ablation for Level 1 would only introduce computational cost and complexity and would not significantly advance the central research problem.

4.7.2 Inter-Molecule Aggregation (Level 2 Ablation)

This is the study’s main ablation test, to determine whether the use of explicit attention-based cross-molecule interaction modeling can meaningfully outperform naive linear pooling. Specifically, the architectural component being swapped out is the Level 2 Set Transformer (SAB + PMA), which will be compared against the mean pooling baseline, which calculates the unweighted arithmetic mean of the component vectors, assuming that the perception of a scent mixture is a linear combination of its single-molecule components, using the formula:

$$\mathbf{z}_{\text{blend}} = \frac{1}{M} \sum_{m=1}^M \mathbf{z}_m \quad (4.8)$$

In addition to the two pooling baselines above, a reproduced pairwise blend model is also included, following the GIN-based architecture Sisson et al. [40] describe as their point-of-comparison model: each molecule in the pair is encoded independently by a Graph Isomorphism Network (GIN), and the two resulting embeddings are concatenated, rather than pooled, before being passed through the same $256 \rightarrow 128 \rightarrow 74$ output head and Asymmetric Focal Loss used elsewhere in this study. Unlike

Sisson et al.’s original evaluation, which used a random 50:50 split, this reproduction is trained and evaluated on the same UMAP-clustered OOD split as the rest of this study’s models, so the comparison isolates the effect of the aggregation method rather than the effect of a different data split.

Everything else will remain constant across the ablation, using the same GAT from Level 1, the same dataset and splitting techniques, training setup, metric suite, and statistical protocol, to ensure that the differences that arise from the ablation are attributed directly to the swaps done. The results of this ablation, combined with the evaluation metrics and significance testing, will directly determine whether this thesis’s central hypothesis, that explicit attention-based interaction modeling meaningfully outperforms naive linear pooling for multi-label blend prediction, is supported.

4.7.3 Attention–Biology Alignment Analysis

To address the third research question, the Pooling by Multihead Attention (PMA) weights the trained Set Transformer assigned to each molecule in a blend are extracted from the test set. The primary test compares blends whose ground-truth labels exhibit suppression against blends without suppression, testing with a Mann–Whitney U test whether suppressing blends show a significantly larger imbalance between the two molecules’ PMA weights than non-suppressing blends. Because this test draws on every suppression case present in the test set rather than a small number of specific molecules, it does not depend on whether any single documented pair happens to survive the leakage-resistant split.

As a secondary, qualitative check, the test set is also searched for three molecule pairs with directly documented receptor-level antagonism from Section 2.4: β -ionone with sulfur-containing malodors [12], L-menthol with 2,3,5-trimethylpyrazine (TMP) [20], and Florhydral with isovaleric acid [21]. Where any of these specific pairs is present, the model’s behavior is reported as consistent with the documented mechanism if the antagonist molecule receives the higher PMA weight and the corresponding masked descriptor is absent from the blend’s predicted labels. Given the small number of candidate pairs, this check is illustrative rather than a basis for a statistical claim, and its absence from the test set does not undermine the primary Mann–Whitney result above.

This analysis overall is treated as exploratory rather than confirmatory: alignment between PMA weight asymmetry and documented antagonism indicates the model’s attention behavior is consistent with known receptor-level mechanisms, but does not constitute independent biological

validation of the underlying mechanism.

4.8 Sustainable Substitute-Finder Prototype

This section will describe the implementation of the sustainable substitute-finder prototype mentioned in Chapter 1 and Section 2.7, in support of UN SDG 12. Specifically, it will rely on the model’s Level 1 GAT encoder, because substitution is inherently a one-on-one single-molecule matching problem, making the Level 1 embedding space more appropriate for this task than the Level 2 space.

A queried SMILES molecule will go through the trained Level 1 GAT encoder, producing a 256-dimensional vector $\mathbf{z}_{\text{query}} \in \mathbb{R}^{256}$, which will then undergo a k -nearest-neighbor cosine similarity search against the pool of candidate substitute embeddings $\mathbf{z}_k$. Cosine similarity was chosen because it measures directional orientation in the latent space rather than vector magnitude, which is what determines qualitative semantic features in learned embedding spaces. This is consistent as well with methodology used by Tom et al. [2025] in their POMMix study. The distance metric formula is as follows:

$$\text{CosineSimilarity}(\mathbf{z}_{\text{query}}, \mathbf{z}_k) = \frac{\mathbf{z}_{\text{query}} \cdot \mathbf{z}_k}{\|\mathbf{z}_{\text{query}}\| \|\mathbf{z}_k\|} \quad (4.9)$$

The candidate pool for substitutes will consist of the single molecules from the Goodscents + Leffingwell single-molecule master dictionary. To ensure viability, this pool is programmatically filtered by cross-referencing the molecules’ SMILES strings and CIDs against published regulatory databases, specifically the IFRA Standards Library and EU REACH exclusion lists. Any compounds flagged as prohibited or restricted, marked as toxic/bioaccumulative, or sourced from endangered flora are permanently dropped from the search index prior to executing any queries.

In order to systematically assess the utility of this proof-of-concept without the need for wet-lab validation, the prototype’s recommendations will be evaluated using the following criteria:

- **Quantitative Evaluation (Precision@5):** The utility of the model will be measured by calculating the Precision at 5 (P@5) metric. This evaluates the percentage of the top five retrieved nearest neighbors that successfully share at least one standardized olfactory descriptor with the queried target molecule.
- **Regulatory and Safety Compliance (Constraint Satisfaction):** A recommendation is only practically useful to a fragrance chemist if it is legally and chemically safe. The prototype guarantees

this industrial utility by restricting its similarity search exclusively to the pre-screened candidate pool described above. By algorithmically masking flagged compounds from the vector space, it ensures that no IFRA-restricted, toxic, bioaccumulative, or endangered compounds can ever be recommended.

By satisfying these two criteria, the prototype demonstrates its applied utility as a sustainable, *in-silico* pre-screening tool. While full physical compatibility (such as Hansen solubility and ODT/OAV matching) and wet-lab confirmation are scoped out and deferred to future work, this structured evaluation ensures the model's outputs are both mathematically accurate and industrially viable.

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